Climate Change, Income Inequality, and Migration in a Spatial Economy

Michael Tueting*

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Abstract

Negative local labor market shocks create strong incentives to migrate, yet low-income households often remain in place. This paper studies how income shapes migration responses to climate change and the resulting welfare effects. Using Brazilian Census data, I show that high-income individuals are systematically more mobile than low-income individuals from the same origin. To interpret this pattern and quantify the impacts of climate change, I develop a dynamic spatial general equilibrium model with monetary migration costs and liquidity constraints, embedded in a two-sector trade framework in which rising temperatures depress agricultural productivity. The quantitative results imply sharply regressive welfare losses: in already hot regions, low-income households experience permanent consumption declines of about 0.6-0.9% per period, with worst-case losses near 3.3%, while richer households are largely insulated. Two mechanisms drive these disparities: climate shocks reduce agricultural wages in hot areas, and monetary migration costs disproportionately burden low-income individuals, limiting their ability to relocate in response. A counterfactual policy that makes low-income individuals as mobile as high-income individuals–modeled as a targeted subsidy–raises low-income individuals' welfare by 24%, offsets 4% of climate losses, and increases aggregate output by relocating labor to more productive regions. Adaptation to climate change depends not only on where productivity shocks occur but also on who can afford to relocate. Consequently, adverse shocks tend to fall disproportionately on those unable to afford relocation.

*Tueting: Swiss Institute for International Economics and Applied Economic Research, University of St.Gallen, Rosenbergstrasse 22 CH-9000 St. Gallen, michael.tueting@unisg.ch. I am grateful for the advice Bruno Caprettini, Doug Gollin, Roland Hodler, Ferdinand Rauch, and Adam Storeygard provided. I also thank Arnaud Costinot, Juanma Castro-Vincenzi, Federico Esposito, Isabella Manelici, Ferdinando Monte, Christoph Moser, Jose Vasquez, workshop participants at Tufts University, and the University of St.Gallen for their helpful comments on early versions of this paper.

1. Introduction

The distribution of economic activity and opportunity across regions is in constant motion. Trade liberalization, technological change, and increasingly climate change are shifting opportunities across sectors and space. In principle, migration allows households to pursue these opportunities. In practice, many individuals remain in place even when the benefits of relocation are substantial. Understanding this gap between incentives and realized mobility is central to both positive and normative economics: it determines who bears the cost of local downturns, how quickly economies reallocate factors in response to shocks, and which policies are effective when place-based fortunes diverge.

This paper examines the role of income in shaping workers' intra-national migration responses to climate change and the resulting aggregate and distributional welfare consequences. Climate change is a particularly salient example of a spatial shock, as rising temperatures affect agricultural productivity and rural incomes unevenly across regions, particularly in low- and middle-income countries. Although migration is frequently considered a key adaptation strategy, empirical evidence indicates weak migration responses to climate shocks by low-income individuals (Cattaneo and Peri, 2016; Bryan, Chowdhury, and Mobarak, 2014). Motivated by this pattern, I develop a dynamic spatial general-equilibrium model that links liquidity constraints to migration, showing how monetary costs associated with migration can suppress mobility among low-income workers facing climate change. By explicitly modeling liquidity constraints, the framework departs from canonical models in which migration costs are assumed to be non-monetary and observed flows reflect optimal choices. This distinction implies that policies aimed at relaxing liquidity constraints, such as subsidizing moving costs, can increase mobility and improve adaptation to climate change.

Brazil is a compelling context for analysis. Spatial inequality is substantial and persistent: the North and Northeast are less wealthy, hotter, and more rural than the South and Southeast. In some northern regions, up to 40 percent of workers are in agriculture, while non-agricultural activities dominate the South and coastal areas. The climatic gradient is also pronounced, with northern regions exhibiting an annual mean temperature of approximately 28°C, which is 10°C higher than that of southern regions.

Two empirical patterns motivate the study. First, among individuals from the same region, higher-income Brazilians exhibit greater mobility than their lower-income counterparts. Analysis of the 2010 Census reveals that migration rates increase steeply with income: individuals earning R$40-50k were nearly twice as likely to have moved between 2005 and 2010 as those earning R$0-10k. This pattern is consistent with the notion that migration requires an upfront monetary cost, such as transportation costs, deposits, and foregone earnings, which impose a disproportionately heavy burden on low-income households. Second, climate change affects agricultural productivity through temperature: yields increase up to 23-24°C, then decline sharply. Under standard warming scenarios, already-hot regions experience productivity losses, while cooler areas gain. An increase in temperatures in the North, therefore, reduces incomes and increases the value of migration while simultaneously constraining the ability to relocate. The result is a powerful asymmetry: precisely where the incentive to migrate is strongest, the means to do so are weakest.

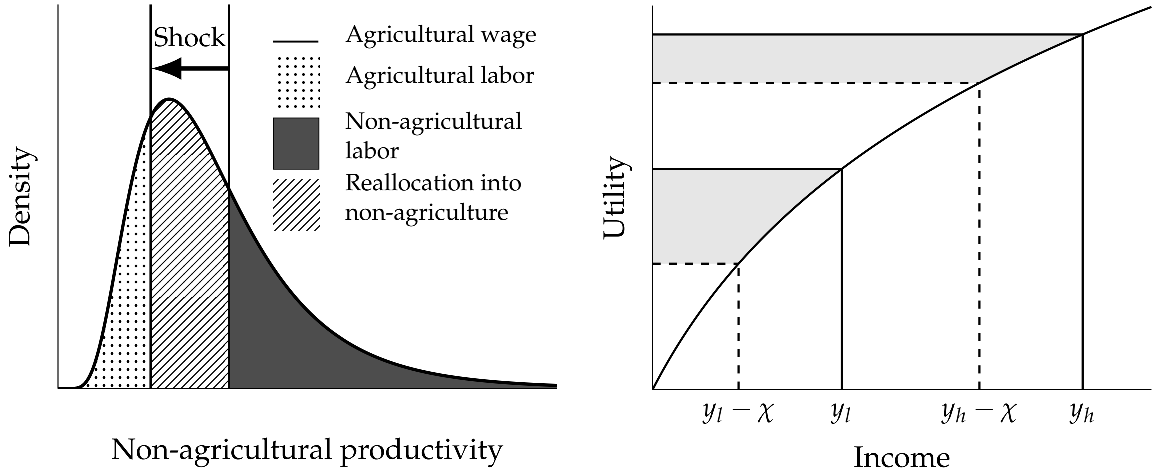
I build a dynamic spatial general-equilibrium model tailored to these facts. (i) Migration involves both monetary and non-monetary costs. Under concave utility, the utility-equivalent burden of a given monetary cost increases as income falls. Consequently, adverse local shocks that reduce wages restrict mobility precisely when wage differentials increase. (ii) Regional income distributions evolve over time: agriculture pays a reservation wage, while non-agricultural sectors offer a continuum of jobs with heterogeneous productivity, each compensating workers according to the marginal product of labor. Workers are matched with non-agricultural jobs and select the most favorable option available. If their non-agricultural wage falls below the agricultural wage, workers opt for employment in agriculture. When climate change reduces a region's agricultural wage, a greater number of workers transition into low-productivity, non-agricultural jobs, which decreases mean income and increases income dispersion.

Panel (a) of Figure I demonstrates this mechanism. Initially, all individuals whose non-agricultural productivity draws exceed the agricultural wage are employed in non-agriculture (dark shaded area). When climate change reduces the local agricultural wage (left arrow), some workers who previously worked in agriculture transition to low-productivity, non-agricultural jobs. If climate change affects regions unevenly, low-productivity workers in the most severely impacted areas have greater incentives to relocate to regions experiencing smaller declines in agricultural productivity. However, when migration requires a fixed monetary investment, the utility-equivalent burden of moving increases as income decreases, thereby reducing migration. Panel (b) of Figure I illustrates this mechanism. For a high-income individual with income y_h, paying a monetary cost χ results in a lower utility-equivalent burden than for a low-income individual (difference in shaded areas). Moreover, workers whose income falls below the monetary cost χ are entirely excluded from migration. In this model, adaptation to climate change through migration depends on where productivity changes occur and who can act on those incentives. Because migration costs are paid in cash, adverse local shocks reduce earnings and the ability to relocate, making them self-reinforcing.

Applying the model to projected climate change from 2010 to 2100, I find that climate damages are strongly regressive. In already-hot regions, low-income households, defined as those earning below R$10,000 (approximately 4,400 USD, near the median wage), experience permanent consumption losses of approximately 0.6 to 0.9 percent per period, with substantially larger losses in later periods. In the most affected regions, losses for low-income households approach 3.3 percent, whereas wealthier households in these areas incur minimal losses. In cooler southern regions, some low-income households benefit as local productivity approaches the agronomic optimum.

I then study a targeted migration subsidy that equalizes the utility-equivalent moving cost across income groups by subsidizing moves for low-income households. This policy increases welfare for low-income households by approximately 24 percent, offsets about 4 percent of their baseline climate losses (3 percent nationally), and increases output by reallocating labor to more productive locations. At a 3 percent annual discount rate, the present-value GDP gains from this policy are 7.9 percent of initial GDP, compared to fiscal costs of 8.7 percent, resulting in a coverage ratio of approximately 90 percent.

FIGURE I
Economic adjustment under productivity shocks.



(A) Distribution of productivity and sectoral labor allocation.

(B) Utility-burden of a monetary migration cost χ.

Notes. Panel (a) illustrates how a negative productivity shock reduces the agricultural reservation wage, reallocating marginal workers into low-productivity non-agricultural jobs. Panel (b) illustrates how a monetary migration cost χ translates into a higher utility-equivalent burden for low-income individuals than for high-income individuals, muting migration even when wage gaps across regions widen.

I contribute to the literature on spatial adjustment, migration, and climate adaptation. My first contribution is to introduce liquidity constraints and monetary migration costs into a quantitative spatial equilibrium framework. These models are widely used to study spatial adjustment in contexts ranging from trade and urban development to regional decline and climate change. In quantitative spatial models, migration costs are typically modeled as utility costs–equivalent to an amenity loss or taste shock–rather than as explicit monetary costs (e.g., Caliendo, Dvorkin, and Parro, 2019; Conte, 2022; Conte, Desmet, Nagy, and Rossi-Hansberg, 2021; Cruz and Rossi-Hansberg, 2024; Cruz, 2024; Desmet and Rossi-Hansberg, 2015; Jedwab, Haslop, Zarate, and Rodriguez Castelan, 2023; Morten and Oliveira, 2024). In the standard framework, observed migration flows are interpreted as first-best responses to migration incentives: costs merely reduce the perceived gains from moving. I instead model migration costs as having both monetary and non-monetary components, and allow for the presence of liquidity constraints. This distinction implies that low-income households may fail to migrate not because they optimally choose to stay, but because they are financially unable to act on incentives to move. Migration outcomes thus reflect constrained rather than optimal choices.

While my application focuses on climate change, the underlying mechanism–migration limited by insufficient liquidity–extends more broadly to spatial downturns and has important implications for optimal policy design. O'Connor (2024), for example, shows that taxing declining regions can raise welfare by encouraging relocation to more productive areas–an implication that holds when migration is inhibited by weak incentives. In my framework, however, when monetary constraints restrict migration, taxation reduces welfare without increasing mobility. Instead, targeted migration subsidies become the efficient policy response.

This addition is not merely a technical extension of the canonical model but a necessary step to align quantitative spatial models with empirical evidence that liquidity constraints limit migration: In Bangladesh, Bryan et al. (2014) find that small, temporary migration subsidies substantially increase seasonal migration and consumption, consistent with binding up-front costs. Cai (2020) shows that credit access raises rural-urban migration in China, and Gazeaud, Mvukiyehe, and Sterck (2023) document similar effects of cash transfers in the Comoros. In the United States, Bastian and Black (2024) demonstrate that expansions of the Earned Income Tax Credit (EITC) increase out-migration from distressed regions, suggesting that even modest income gains can enable efficient relocation. Across these settings, individuals appear immobile not for a lack of incentives, but for a lack of means.

Earlier structural papers that incorporate liquidity constraints include Kennan and Walker (2011), Bryan et al. (2014), Aron-Dine (2024), and Kleemans (2015), who model migration under risk and liquidity constraints within partial-equilibrium, household-level frameworks. These studies provide key microfoundations for understanding how liquidity constraints shape individual migration decisions, but they abstract from the general equilibrium feedbacks that arise when migration affects local incomes, prices, and regional productivity. My paper builds on these insights by embedding such frictions within a fully fledged quantitative spatial general equilibrium model, where income both affects and is affected by mobility.

Beyond this theoretical contribution, my second contribution is to the literature on climate change and climate-induced migration. A large empirical literature documents that temperatures rising above an agronomic optimum reduce agricultural productivity and income (e.g., Schlenker and Roberts, 2009; Lobell and Gourdji, 2012; Burke, Hsiang, and Miguel, 2015; Feng, Krueger, and Oppenheimer, 2010). To formalize this mechanism, I integrate temperature-driven productivity shocks into a spatial general equilibrium model with heterogeneous land endowments, building on recent advances in quantitative trade and spatial economics (e.g., Costinot et al., 2016; Gouel and Laborde, 2021; Sotelo, 2020; Pellegrina, 2022; Conte, 2022; Farrokhi and Pellegrina, 2023; Pellegrina and Sotelo, 2021). While this literature extends the Ricardian trade framework of Eaton and Kortum (2002) to agriculture to model comparative advantage and land-use allocation, it typically abstracts from household-level migration frictions. In the context of climate-induced migration, there is ample evidence that migration responses to temperature shocks are often muted in low-income settings (Bastos, Busso, and Miller, 2013; Brunel and Liu, 2025; Bryan, Chowdhury, and Mobarak, 2014; Bazzi, 2017). My framework bridges this gap by linking climate-induced productivity changes to internal migration decisions under monetary and liquidity constraints, thereby connecting spatial production patterns to household mobility.

Finally, my findings reinforce existing evidence that migration costs hinder efficient labor reallocation and reduce aggregate output (Morten and Oliveira, 2024; Pellegrina and Sotelo, 2021; Bryan and Morten, 2019), but reveal a new dimension of heterogeneity: liquidity constraints disproportionately bind for low-income workers. As a result, targeted migration subsidies that relax these constraints can enhance equity and aggregate efficiency by enabling mobility toward high-productivity regions.

My contributions are threefold. First, I empirically document a pronounced within-origin income gradient in internal migration in Brazil, considering both migration rates and migration distances. Second, I develop a dynamic spatial general equilibrium model that incorporates monetary migration costs and regional income distributions that evolve over time in response to migration and climate change. Third, I demonstrate that climate-induced income losses disproportionately affect low-income households and that a targeted migration subsidy significantly improves welfare for these households while increasing aggregate output.

The rest of the paper is organized as follows. Section 2 describes data and measurement. Section 3 presents motivating evidence on migration patterns, and section 4 estimates the impact of rising temperatures on productivity. Section 5 presents the theoretical framework. Section 6 details the calibration. Section 7 quantifies welfare losses from climate change and documents the heterogeneity across regions and the income distribution. Section 8 studies an income-targeted migration subsidy and its macroeconomic incidence. Section 9 discusses broader policy implications, and section 10 concludes.

2. Data and measurement

The paper uses three main data inputs: information on employment patterns, household incomes, and migration flows; annual agricultural production figures paired with temperature records; and climate model projections of regional temperatures through 2100. In the following, I describe the unit of observation, the primary data sources, and the measurement of key variables used throughout the analysis.

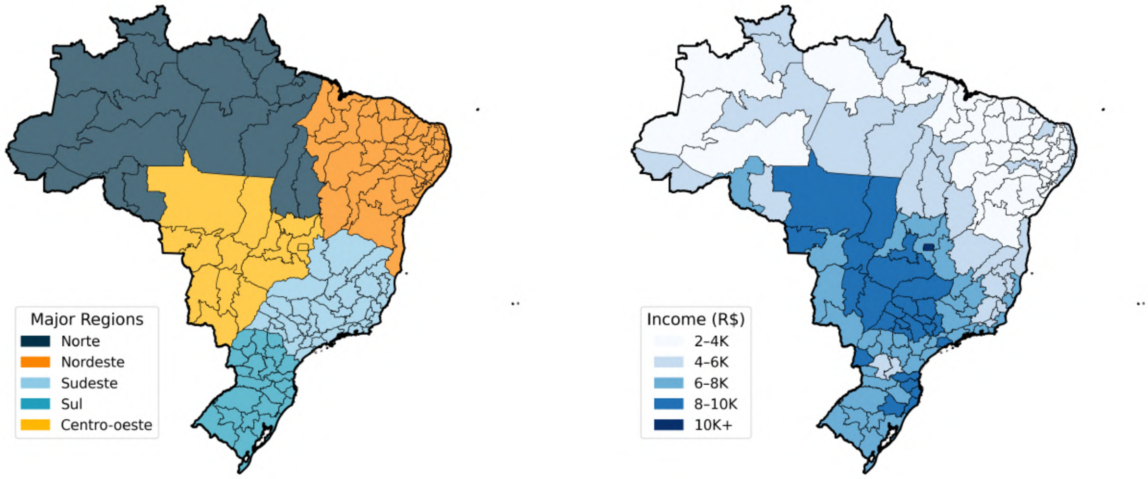
Unit of observation The unit of observation is the mesoregion. A mesoregion is a subnational statistical unit in Brazil, which is defined by the Brazilian Institute of Geography and Statistics (IBGE). There are 137 mesoregions, which are grouped into five larger geographic regions: Norte, Nordeste, Centro-Oeste, Sudeste, and Sul. While mesoregions are not administrative units, they offer the advantage of being stable units over time (since 1989) and ease inter-temporal comparisons. They have also been frequently used in related literature (Gollin and Wolfersberger, 2024; Pellegrina and Sotelo, 2021). Figure II (left-panel) shows the mesoregions of Brazil (dark lines) and the respective major geographical regions.

Socioeconomic data Socioeconomic data on migration flows, employment patterns, and household incomes come from the 2010 Brazilian Census of the Population (*Censo Demográfico*). The sample consists of Brazilian-born individuals aged 25-64 with positive income and non-missing information on occupation, place of residence in 2005, race, education, marital status, and gender, resulting in approximately 6.2 million observations, which correspond to about 60.3 million individuals in the weighted population.[1]

Migration flows are constructed from reported municipalities of residence in 2005 and 2010, aggregated into mesoregion-to-mesoregion flows. Table I presents summary statistics on the fraction of respondents who, in 2005 and 2010, were living in the same mesoregion, a different mesoregion within the same state, and a different state.

[1] The Census extract includes around 20 million observations. Restricting the sample to the working-age population (25-64) reduces the sample by approximately 10 million observations. Dropping those with missing occupation codes (notably, this includes individuals not in the labor force) reduces the sample to 6.5 million observations. The other restrictions reduce the sample size only minimally to the final dataset of 6.2 million observations.

FIGURE II
Regions of Brazil & agricultural income



(A) Meso & Major regions of Brazil

(B) Average annual income of agricultural workers

Notes. Panel (a) shows the five major geographic regions of Brazil: Norte, Nordeste, Centro-Oeste, Sudeste, and Sul, as well as the 137 mesoregions defined by the Brazilian Institute of Geography and Statistics (IBGE). Mesoregion boundaries are marked with thin black lines, and shapefiles were obtained from the IBGE's official geographic database. Panel (b) illustrates the average annual income of agricultural workers in each mesoregion for 2010, based on microdata from the 2010 Brazilian Population Census (Censo Demográfico). The sample includes Brazilian-born individuals aged 25 to 64 who reported positive income and provided valid demographic, occupational, and geographic information. Income refers to total annualized gross earnings from employment in crop cultivation, reported in 2010 Brazilian Reais (R$). The 2010 exchange rate was approximately 1 USD to R$2.3. Crop cultivation occupations are classified using the 5-digit occupation code. Further information on the sample and occupation code mapping is available in appendix A.1.

TABLE I
Patterns in migration flows

Residence	2005-2010
Same mesoregion	95.6%
Different mesoregion, same state	1.8%
Different state	2.6%

Notes. The table presents the proportion of individuals who lived in the same mesoregion, a different mesoregion within the same state, or a different state between 2005 and 2010. This measurement utilizes microdata from the 2010 Brazilian Population Census (Censo Demográfico) and includes Brazilian-born individuals aged 25 to 64 with positive reported income and complete demographic, occupational, and geographic information. Migration flows are identified by mapping municipality-of-residence codes from 2005 and 2010 to mesoregion-to-mesoregion transitions, following the classification of the Brazilian Institute of Geography and Statistics (IBGE). Additional information on the sample and migration flow construction is available in Appendix A.1.

I map Census occupation codes to those working in crop cultivation and non-crop related occupations. Throughout the paper, I refer to the former as "agriculture" and the latter as "non-agriculture". This concept of agriculture is narrower than the agricultural sector in national accounts, which includes activities such as livestock, forestry, and fishing. However, to align the model with my measure of agricultural productivity (based solely on crop production), I focus on crop cultivation. Sectoral employment shares are then measured using the share of workers employed in crop cultivation and non-crop activities. Income is calculated as the annualized total gross monthly income reported in the Census. The average annual income for workers in crop cultivation in 2010 is presented in Figure II (right panel) in 2010 Brazilian Reais (R$).

Appendix A.1 shows summary statistics on the sample of workers and the mapping of occupation codes to workers employed in crop cultivation.

Agricultural production Data on agricultural production are sourced from the Produção Agrícola Municipal (PAM) of the Brazilian Institute of Geography and Statistics (IBGE), which provides municipal-level statistics on more than 60 crops from 1994 to 2014. Municipal observations are aggregated to the mesoregion level, resulting in annual measures of crop-specific area planted, physical output, and production value. Total agricultural land is defined as the cumulative area planted with crops, and this measure is used to construct mesoregion-level land-use statistics from 1994 onward.

Measured agricultural productivity is defined as the value of crop output per unit of land at constant prices.[2] By fixing prices at nationwide reference levels, differences in measured productivity reflect variation in physical yields rather than price heterogeneity across crops, regions, or time. This approach follows standard practice in the literature (e.g., Gollin, Hansen, and Wingender, 2021; Restuccia, Yang, and Zhu, 2008). In both the empirical analysis and the calibrated model, measured agricultural productivity is adjusted for land use intensity following Costinot, Donaldson, and Smith (2016). Intuitively, land quality varies within each region, and farmers typically cultivate the most productive fields first. As a result, the average (unconditional) productivity of land in a region is generally lower than the measured productivity due to positive selection of land into cultivation.[3] Appendix B.2 outlines the procedure and provides further summary statistics.

Temperature data Temperature data come from the Copernicus Climate Change Service (C3S) CMIP6 archive. I use monthly near-surface air temperature from the CMCC-ESM2 Earth system model, available on a 1° x 1° grid, covering the historical period through 2014 and future projections under the SSP2-4.5 and SSP5-8.5 scenarios. Temperatures are converted from Kelvin to Celsius, aggregated to yearly means, and averaged to the mesoregion level using land-area weights. The historical data provide annual mesoregion level means from 1994 to 2014, which I match to PAM, and the projections extend through 2100. For the projections, I compute five-year averages and express anomalies relative to 2010.

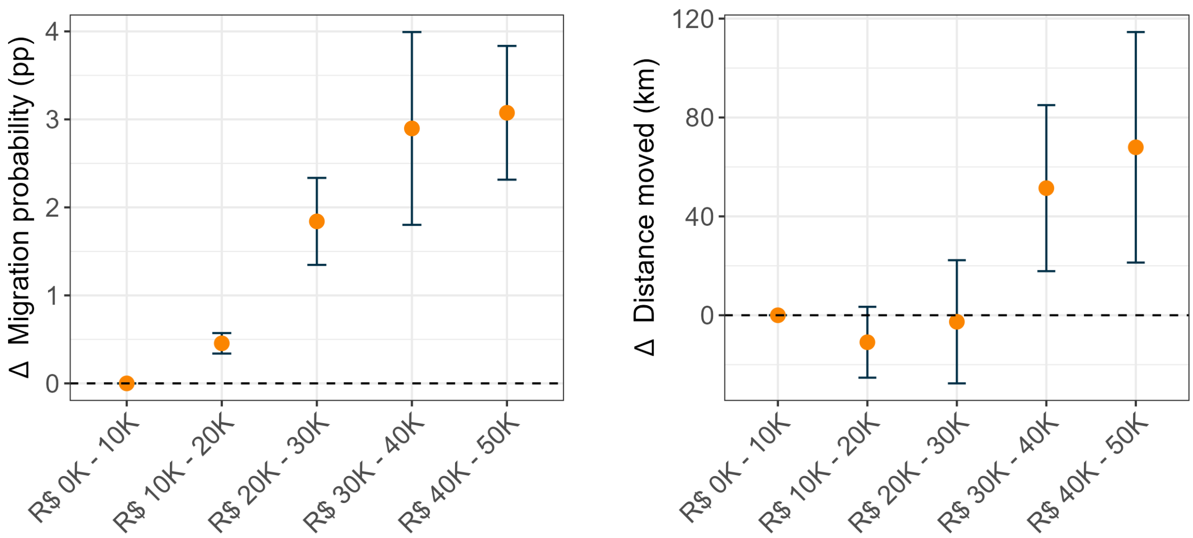
[2]For each crop c, I compute a nationwide reference price $P_c^* = (\sum_i \sum_t V_{c,it})/(\sum_i \sum_t Q_{c,it})$, where $V_{c,it}$ is the nominal value of production and $Q_{c,it}$ is the physical quantity in region i and year t, using all years between 1994 and 2014. Constant price output in region i and year t is then $\widetilde{V}_{it} = \sum_c P_c^* Q_{c,it}$, and measured productivity is $\tilde{A}_{it} = \widetilde{V}_{it}/(\sum_c X_{c,it})$, where $X_{c,it}$ is the land area planted with crop c.

[3]For example, Rio de Janeiro ranks second based on measured agricultural productivity; however, its share of land under cultivation is only around 2% of its land area. Without adjusting for land-use intensity, this ranking reflects selection into a small set of highly productive plots rather than a high region-wide average.

3. Motivation: Who migrates?

In many low- and middle-income countries, households live close to subsistence, with limited savings and limited access to credit. In Brazil, 8.4% of the population lived below the \$3/day poverty line in 2010,[4] with poverty concentrated in the North and Northeast. For these households, financing large one-time expenses such as migration is often prohibitive. Liquidity constraints, therefore, predict that low-income households migrate less, despite potentially stronger incentives to escape structural poverty. Consistent with this perspective, I document that in Brazil, higher-income individuals are substantially more likely to migrate than low-income individuals and tend to move farther than less wealthy individuals from the same origin.

FIGURE III
Migration & income



(A) Difference in migration rates by income (B) Difference in distance of moves by income

Notes. This figure presents coefficients from regressions that relate migration outcomes between 2005 and 2010 to predicted counterfactual 2010 income bins. Counterfactual incomes are estimated using region-specific Mincer regressions of 2010 log income on age, age squared, gender, race, marital status, and education, based on the sample of non-movers. These estimates are then used to predict what 2005 residents would have earned in 2010 had they remained in their original locations. Predicted incomes are categorized into five R\$10,000 bins up to R\$50,000, with the lowest bin (0-10,000 R\$) omitted. The distribution of individuals across bins is as follows: 47% in 0-10K, 38% in 10-20K, 7.8% in 20-30K, 4.2% in 30-40K, and 2.2% in 40-50K. Panel (a) displays percentage-point differences in the probability of migrating (defined as changing mesoregion) relative to the omitted bin. Panel (b) shows differences in the distance moved (in kilometers), conditional on migration, relative to the omitted bin. Vertical lines indicate the 95% confidence intervals of the point estimates. Standard errors in panel (a) are clustered by origin mesoregion, while those in panel (b) are clustered by both origin and destination. The sample consists of Brazilian-born individuals aged 25 to 64 who report positive income and complete demographic and occupational information.

To quantify this relationship, I construct counterfactual incomes representing what individuals would have earned in 2010 had they remained in their 2005 mesoregion. These pre-

[4]Measured at 2011 PPP according to the World Bank's Poverty and Inequality Platform. https://pip.worldbank.org

dicted incomes are derived from region-specific Mincer regressions of 2010 log income on age, age squared, gender, race, marital status, and education. The resulting fitted values are interpreted as earnings capacity in the absence of migration, assuming that (i) conditional on observables, the 2010 wage structure reflects relative opportunities for 2005 residents, and (ii) the small migrant share, approximately 5%, does not significantly affect local wages.

Predicted incomes are grouped into five 10,000 R$ bins up to 50,000 R$, which I then use to estimate how migration probabilities and the average distance of actual moves vary across the income distribution. The estimating equation is:

$$Y_{ijt} = \sum_{b=2}^{5} \beta_b \, \mathbf{1}\{\text{Pred. Inc.}_{ijt} \in b\} + \mu_j + \varepsilon_{ijt},$$

where i indexes individuals, j indexes the 2005 mesoregion of residence, and t indexes the 2010 cross-section (with outcomes measured over 2005-2010). Y_{ijt} is the individual outcome, μ_j are mesoregion fixed effects, and income bin $b = 1$ (R$0-10,000) is the omitted category.

Panel (a) of Figure III defines Y_{ijt} as a binary indicator of migration between 2005 and 2010, with β_b denoting percentage-point differences in migration probability relative to the omitted category. In panel (b), Y_{ijt} measures the distance moved in kilometers between 2005 and 2010, estimated from the sample of movers only. Here, β_b indicates differences in kilometers relative to the omitted category.

Panel (a) demonstrates a pronounced upward gradient. Relative to the omitted category, migration probabilities increase by approximately 0.5 percentage points for R$10-20,000, 1.86 percentage points for R$20-30,000, and about three percentage points for both R$30-40,000 and R$40-50,000. The average emigration rate for individuals in the lowest income group is 3.93%. This indicates that individuals in the highest income groups are nearly twice as likely to have migrated between 2005 and 2010.

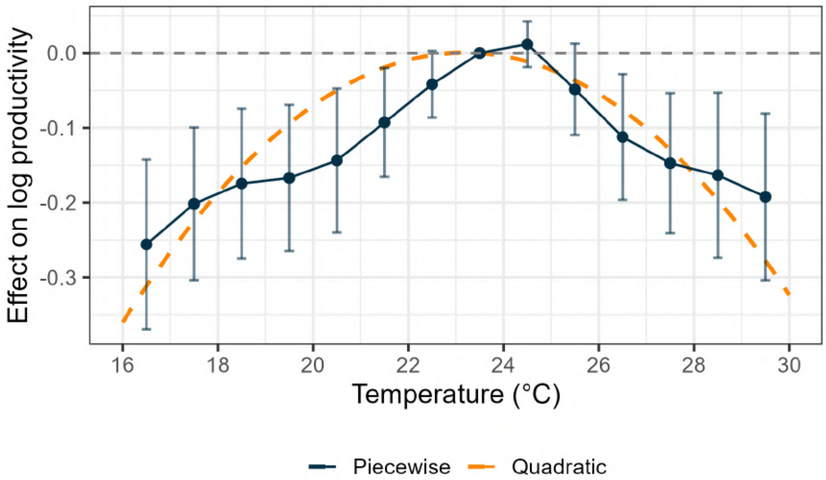
Panel (b) indicates no statistically significant difference in distance moved for the two lower-income categories. However, among higher predicted incomes, movers travel substantially farther: approximately 50 km more for R$30-40,000 and about 70 km more for R$40-50,000. The average distance moved by the few movers in the lowest income group is 700 km, suggesting that the highest income groups move, on average, 10% farther.

Collectively, these results support the interpretation that liquidity constraints influence both the likelihood of migration and the distance moved. Low-income households face binding upfront cash requirements for transport and resettlement, which exclude many high-cost, high-return destinations at the extensive margin. Even when costs are payable, fixed costs represent a larger utility-equivalent sacrifice for these households under concave utility, reducing responsiveness to opportunity differences at the intensive margin. As distance increases costs, these constraints render long-distance moves feasible primarily for higher-income individuals, thereby reinforcing spatial inequality and limiting migration as an adaptation strategy for low-income groups.

4. The Impact of Climate Change on Agricultural Productivity

To establish the relationship between temperature and agricultural productivity in Brazil, I estimate how yields change with temperature. This relationship is central to the analysis because it determines how climate change influences rural incomes and subsequently affects migration incentives.

FIGURE IV
Estimated relationship between temperature and agricultural productivity.



Notes. The figure presents the estimated relationship between temperature and fundamental agricultural productivity, measured as constant-price output per hectare of land adjusted for land use intensity, in Brazilian mesoregions from 1994 to 2014. The solid blue line represents binned model estimates, and the vertical bars indicate 95% confidence intervals calculated using standard errors clustered at the mesoregion level to account for serial correlation within mesoregions in temperature and productivity over time. The dashed orange line depicts quadratic model estimates. Both models incorporate mesoregion fixed effects and region-specific cubic time trends.

Following Costinot, Donaldson, and Smith (2016), I adjust measured productivity (constant price output per hectare of land) for land use intensity to obtain a measure of fundamental productivity. Fundamental productivity is defined as $A_{it} = \tilde{A}_{it} \cdot \zeta_{it}^{1/\theta}$. In this expression, $\tilde{A}_{it}$ denotes measured productivity, ζ_{it} represents the share of land allocated to agriculture, and θ captures the degree of heterogeneity in land quality. The parameter θ is set to 2, reflecting the midpoint of available estimates (Sotelo, 2020; Costinot, Donaldson, and Smith, 2016). To estimate the impact of temperature on fundamental productivity, I exploit variation in average temperature within mesoregions over time. I estimate a piecewise linear and a quadratic specification of the following form:

$$\ln A_{it} = \sum_{b \in \mathcal{B} \setminus b^{\star}} \beta_b \, \mathbf{1}\{TG_{it} \in b\} + \mu_i + \sum_{p=1}^{3} \tau_{ip} \, t^p + \varepsilon_{it} \qquad \text{(piecewise)}$$

$$\ln A_{it} = \beta_1 \, TG_{it} + \beta_2 \, TG_{it}^2 + \mu_i + \sum_{p=1}^{3} \tau_{ip} \, t^p + \varepsilon_{it} \qquad \text{(quadratic)}$$

where TG_{it} denotes the annual mean temperature of mesoregion i in year t, μ_i are mesoregion fixed effects, and $\{\tau_{ip}\}_{p=1}^{3}$ allows for region-specific linear, quadratic, and cubic time trends. The piecewise specification uses 1°C bins, with $[23, 24)$ as the omitted category.

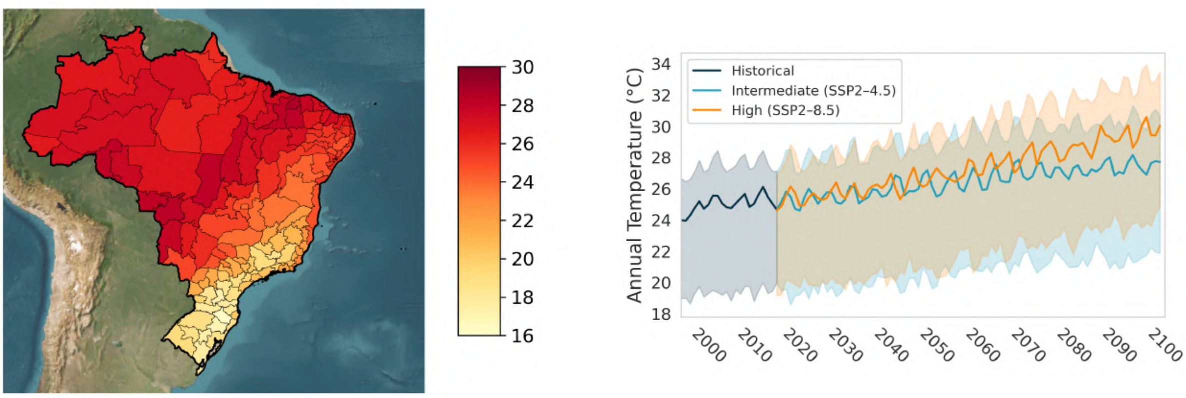
Figure IV presents the estimated relationship between temperature and productivity. Both the binned and quadratic models indicate a concave relationship: productivity increases with temperature up to around 23-24°C and declines sharply thereafter.

The quadratic model estimates an optimal temperature of about 23.5°C, with a 95% confidence interval of [22.8, 24.1]°C.[5]

This nonlinear relationship suggests that the effects of warming vary across regions. Cooler areas, such as those in the South, could experience productivity gains from moderate temperature increases. In contrast, hotter regions, particularly in the North and Northeast, are already operating on the downward-sloping segment of the curve and are thus particularly vulnerable to further temperature increases.

Brazil exhibits a pronounced upward temperature trend. Figure V displays the average annual temperature by mesoregion in 2010 (left panel) and the historical and projected annual mean temperatures under the SSP2-4.5 and SSP5-8.5 scenarios (right panel). In panel b, the solid line represents the median regions' temperature for each scenario, with shaded areas denoting the 10th and 90th percentiles. By 2100, average regional temperatures are projected to increase by 2.9°C under SSP2-4.5 and by 4.1°C under SSP5-8.5, with some regions surpassing 30°C in the high-emission scenario.

FIGURE V
Temperature in Brazil



(A) Average annual temperature in 2010

(B) Projected temperature under SSP2-4.5 and SSP5-8.5

Notes. Panel (a) presents the average annual temperature for 2010, derived from the Copernicus Climate Change Service (C3S) CMIP6 archive. Panel (b) displays the median regional annual temperatures in Brazil from 1994 to 2100 under the SSP2-4.5 and SSP5-8.5 scenarios, simulated with the CMCC-ESM2 Earth System Model. Temperature data are provided on a 1° × 1° grid, converted from Kelvin to Celsius, aggregated to annual means, and averaged at the mesoregion level using land-area weights. The historical period spans 1994 to 2014, and future projections extend to 2100. For visualization, projected values are presented as five-year averages, with shaded areas representing the 10th and 90th percentiles across mesoregions.

[5]Under the quadratic specification, the optimum is $\hat{T}^* = -\hat{\beta}_1/(2\hat{\beta}_2)$ (with $\hat{\beta}_2 < 0$); the interval is computed via the delta method using the estimated variance-covariance matrix of the β coefficients.

Consequently, many regions are expected to enter temperature ranges where productivity declines sharply, intensifying challenges for agricultural workers and rural markets. Linking agricultural productivity to incomes, I estimate that a 1% increase in agricultural productivity raises agricultural wages by 0.21% (see Appendix B.2 for details).

5. Theoretical framework

I develop a quantitative spatial model to study welfare, migration, and income dynamics in response to climate change. The model consists of two interconnected components: a dynamic discrete-choice problem and a static trade equilibrium.

These components interact as follows: the static trade model determines wages, prices, and consumption levels *conditional on a given allocation of labor and regional fundamentals*. The dynamic component then solves for the optimal reallocation of labor over time, *based on the expected path of real incomes implied by the static equilibrium.*

Dynamic problem: In the dynamic household problem, households take the structure of the economy as given and make forward-looking migration decisions that shape migration flows between regions over time. The framework builds upon the seminal work of Caliendo, Dvorkin, and Parro (2019), which represents the current state of the art in this literature.

Migration costs are modeled as origin-destination-specific monetary costs, in addition to a non-monetary psychic cost of leaving "home". With concave utility, the monetary cost imposes a larger burden on low-income households. In contrast, Caliendo et al. (2019) assume income-invariant utility costs tied to origin-sector and destination-sector pairs. This distinction has significant implications. In the model of Caliendo et al. (2019), mobility is unaffected by income shocks, conditional on constant relative utility differences across space. In the model presented here, lower incomes increase the effective burden of moving and reduce mobility. This mechanism captures an important channel of climate change: falling incomes make migration more difficult, particularly for low-income workers.

Static sub-problem: The second component is a static trade model with two sectors and multiple regions. Unlike standard quantitative spatial models (see Redding and Rossi-Hansberg, 2017), this framework embeds region-specific, continuous income distributions that evolve endogenously over time. Within each region, agriculture offers a single market-clearing wage that serves as a reservation wage. Non-agricultural employment features a continuum of jobs with heterogeneous productivity levels, with wages equal to the marginal product of labor. Workers draw a productivity realization in this sector; if the implied wage is below the agricultural reservation wage, they work in agriculture instead. This mechanism parallels Lucas (1978) and Luttmer (2007), where individuals draw entrepreneurial productivity and choose between entrepreneurship and wage work.

Climate shocks that reduce agricultural productivity lower the reservation wage, which pushes more workers into low-productivity, non-agricultural jobs. The result is higher inequality and lower average income. Climate change affects the model in two ways: by shifting agricultural productivity and regional income distributions, and by increasing the income-dependent utility burden of migration costs.

The remainder of this section is structured as follows. First, the dynamic component is described, with emphasis on the novel feature introduced into the baseline model of Caliendo et al. (2019): monetary migration costs. Second, the static trade model is presented, in which climate change interacts with income distribution both across and within regions. Online Appendix C provides the full derivations, and Online Appendix F outlines how the full model is solved in practice.

5.1. Dynamic component

As discussed above, many households in Brazil live close to subsistence, with a significant proportion below the $3 per day poverty line. For these households, financing large, one-time expenditures, such as migration, is prohibitive. I therefore abstract from asset accumulation and model migration decisions as depending solely on income realizations and migration costs.

5.1.1. Setup and timing

The economy consists of a finite set of regions $j, i \in \mathcal{S}$, each populated by a mass of households.[6] Time is discrete, $t \in \{1, \ldots, T\}$. Households may migrate at the end of a period, but have to pay a monetary cost $\chi^{j,i}$ as well as a non-monetary cost to do so. The future is discounted at a rate $\beta > 0$, and households consume their entire income net of migration costs.

The ability to pay the monetary migration cost depends on individual income. At the start of each period, a household in region j draws from the region's non-agricultural productivity distribution. If the draw implies a wage below the agricultural wage, the household chooses to work in agriculture. Thus, the regional income distribution at time t, f_t^i, emerges endogenously from the agricultural wage and the non-agricultural productivity distribution. Productivity draws follow a Calvo-style process. With probability ρ, the household retains its previous draw and thus its relative position in the non-agricultural distribution. With probability $1 - \rho$, the household draws anew from the uniform distribution on $(0, 1)$. The parameter ρ governs **income persistence**.[7]

Households also draw idiosyncratic location preferences. They observe the full wage distributions in all regions and, therefore, know the expected income prospects in each location. The only uncertainty pertains to their own future wage realizations within those distributions and future preference shocks.

5.1.2. Instantaneous utility

ASSUMPTION 1: *Agents have logarithmic preferences.*

An individual with income y in region j who chooses to migrate to region i at time t derives utility

$$U(C_t^{j,i}(y)) = \log C_t^{j,i}(y),$$

[6] j denotes the origin and i the destination.

[7] This also implies persistence in sectoral choices. A worker with a low initial non-agricultural productivity draw is likely to stay in agriculture for multiple periods for $\rho > 0$.

where consumption equals disposable real income:[8]

$$C_t^{j,i}(y) = \frac{y_t^j - \chi^{j,i}}{P_t^j}.$$

Here, y_t^j denotes an individual's income in region j, $\chi^{j,i}$ the monetary migration cost from j to i, and P_t^j the regional price index. Migration costs capture expenditures such as transportation and housing deposits.

5.1.3. Utility equivalent costs

Income-invariant monetary costs imply income-dependent utility costs. Let $C_t^j(y) = y_t^j / P_t^j$ denote consumption if a household in j does not migrate. The utility-equivalent cost of migration is then

$$\tilde{\tau}^{j,i}(y) \equiv \log\left(\frac{y_t^j}{y_t^j - \chi^{j,i}}\right),$$

which follows from $U(C_t^{j,i}(y)) = U(C_t^j(y)) - \tilde{\tau}^{j,i}(y)$, with $U(C_t^j(y)) = \log(y_t^j/P_t^j)$. This expression is well defined only for $y_t^j > \chi^{j,i}$; whenever $y_t^j \leq \chi^{j,i}$, I set $\tilde{\tau}^{j,i}(y) = +\infty$ (migration is infeasible).

COROLLARY 1: *The utility burden $\tilde{\tau}^{j,i}(y)$ is decreasing with income.*

In other words, monetary costs map into larger utility losses for low-income households. This contrasts with Caliendo, Dvorkin, and Parro (2019), where relocation costs are specified directly in utility terms and remain independent of income.

In addition, I allow for non-monetary costs κ^j of leaving "home". Composite migration costs are

$$\tau^{j,i}(y) = \tilde{\tau}^{j,i}(y) + \mathbf{1}(i \neq j) \cdot \kappa^j.$$

5.1.4. Expected lifetime utility

Let $v_t^j(r)$ denote the lifetime utility of a household residing in region j at time t, conditional on a non-agricultural productivity draw corresponding to rank $r \in (0,1)$ in the local productivity distribution. The associated income is $y_r^j = (F_t^j)^{-1}(r)$, where $(F_t^j)^{-1}$ denotes the inverse CDF of the regional income distribution that arises endogenously from the agricultural reservation wage and the distribution of non-agricultural productivity draws.

Households consume all income net of migration costs and choose the region in which they want to start the next period to maximize expected lifetime utility. The dynamic problem is

$$v_t^j(r) = U(y_r^j) + \max_{i \in \mathcal{S}} \left\{ -\ \tau^{j,i}(y_r^j) + \nu\, v_t^i + \beta\Big[\rho\, \mathbb{E}_t[v_{t+1}^i(r)] + (1-\rho) \int_0^1 E[v_{t+1}^i(r')]dr'\Big] \right\}.$$

where $\tau^{j,i}(y_r^j)$ denotes the effective migration cost from j to i, which depends on income; v_t^i represents an idiosyncratic preference shock for region i; and ν scales the dispersion of these

[8] This follows from the static block presented below.

shocks. The term in brackets captures the expected continuation value, accounting for income persistence governed by ρ and the possibility of a new wage draw. With a probability of ρ, the previous rank is retained, so that the lifetime utility of starting in i in $t+1$ is $E[v^i_{t+1}(r)]$, and with a probability of $1-\rho$, the lifetime utility is the average over all ranks $\int_0^1 E[v^i_{t+1}(r')]dr'$.

Assumption 2. *Preference shocks v^i_t are i.i.d. across i and t and follow a Type-I Extreme Value (Gumbel) distribution with location 0 and unit scale. The dispersion parameter $\nu > 0$ scales these shocks.*

Under Assumption 2, the dynamic discrete choice problem admits the standard logit closed-form expression for expected lifetime utility (McFadden, 1974). Let $V^j_t(r) \equiv \mathbb{E}[v^j_t(r)]$ and $\bar{V}^i_t \equiv \int_0^1 V^i_t(r')\,dr'$. Then, the expected lifetime utility of a worker with rank r in the non-agricultural productivity distribution (associated with income y_r) currently located in region j is:

$$V^j_t(r) = U(C^j_t(y_r)) \;+\; \nu \log \sum_{i\in\mathcal{S}} \exp\Big(\beta\Big[\rho\, V^i_{t+1}(r) + (1-\rho)\,\bar{V}^i_{t+1}\Big] - \tau^{j,i}(y_r)\Big)^{1/\nu}. \tag{1}$$

The first term is the period-t utility from consuming at origin j with current income rank r. The second term, $\nu \log \sum_{i\in\mathcal{S}} \exp(\cdot)^{1/\nu}$, is the *inclusive value*: it summarizes the continuation value of the migration choice set *before* idiosyncratic tastes and next period's productivity draw are realized.[9]

5.1.5. Migration flows and labor dynamics

Under Assumption 2, destination choice probabilities take a logit form. For a household in j with rank $r \in (0,1)$ (income $y^j_r = (F^j_t)^{-1}(r)$), the probability of migrating to i at the end of period t is

$$\mu^{j,i}_t(r) \;=\; \frac{\exp\Big(\big[\beta(\rho V^i_{t+1}(r) + (1-\rho)\bar{V}^i_{t+1}) - \tau^{j,i}(y^j_r)\big]/\nu\Big)}{\sum_{k\in\mathcal{S}} \exp\Big(\big[\beta(\rho V^k_{t+1}(r) + (1-\rho)\bar{V}^k_{t+1}) - \tau^{j,k}(y^j_r)\big]/\nu\Big)}. \tag{2}$$

With a large mass of households, $\mu^{j,i}_t(r)$ is the share of type-r households in j that relocate to i.

Intuitively, households are more likely to move to destinations that offer higher expected future payoffs and are accessible at low migration costs. Monetary costs depress mobility among low-income individuals via (i) an intensive margin, since $\partial\tau^{j,i}(y^j_r)/\partial y^j_r < 0$, and (ii) an extensive margin, since migration is infeasible when $\chi^{j,i} \geq y^j_r$ (so $\tau^{j,i}(y^j_r) = \infty$). This helps explain why poorer households are more likely to stay, even when the returns to moving are large. Persistence also matters: with probability ρ households keep their rank and weigh destinations favorable to their type; with probability $1-\rho$ they redraw and weigh average prospects. The logit scale ν governs choice dispersion: as $\nu \to 0$, choices become nearly deterministic, and flows concentrate on the destination that maximizes $\beta(\rho V^k_{t+1}(r) + (1-\rho)\bar{V}^k_{t+1}) - \tau^{j,k}(y^j_r)$; as $\nu \to \infty$, probabilities flatten across feasible destinations, while infeasible options remain at zero.

[9]Formally, $\nu \log \sum_{i\in\mathcal{S}} \exp(\cdot)^{1/\nu} = \mathbb{E}_{v_t}\Big[\max_{i\in\mathcal{S}}\Big\{\beta\Big[\rho\, V^i_{t+1}(r) + (1-\rho)\,\bar{V}^i_{t+1}\Big] - \tau^{j,i}(y_r) + \nu\, v^i_t\Big\}\Big]$.

Aggregating over the origin's rank distribution (uniform on $(0,1)$) yields aggregate migration probabilities from j to i,

$$\mu_t^{j,i} = \int_0^1 \mu_t^{j,i}(r)\, dr, \tag{3}$$

which is equivalent $\mu_t^{j,i} = \int \mu_t^{j,i}(F_t^j(y))\, f_t^j(y)\, dy$ by the change of variables $r = F_t^j(y)$, where $f_t^j(y)$ is the income density in j.[10]

Aggregate labor stocks evolve according to

$$L_{t+1}^i = \sum_{j \in \mathcal{S}} \mu_t^{j,i} L_t^j, \tag{4}$$

which serve as a key input to the static sub-problem presented next.

5.2. Static trade model

The static block determines wages, prices, land use, and income distributions given the spatial allocation of labor. In the following, I present the minimal setup. The full model and derivations are described in Appendix C.

5.2.1. Setup

Preferences. Households consume an agricultural composite $C_t^{i;a}$ and a non-agricultural good $C_t^{i;n}$ with Cobb-Douglas utility:

$$C_t^i = \frac{(C_t^{i;a})^\gamma (C_t^{i;n})^{1-\gamma}}{\gamma^\gamma (1-\gamma)^{1-\gamma}},$$

where $\gamma \in (0,1)$ is the expenditure share on agriculture. The agricultural composite is a CES aggregate of regional varieties

$$C_t^{i;a} = \left(\sum_{j \in \mathcal{S}} (\psi^j)^{1/\epsilon} (c_t^{j,i;a})^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}},$$

with Armington elasticity ϵ and preference weights ψ^j (normalized to sum to one). Here $c_t^{j,i;a}$ is the consumption of the variety from origin j. Transporting variety $c_t^{j;a}$ to i incurs iceberg trade costs. The non-agricultural good is freely traded and serves as the numeraire.

[10]For tractability, I impose a re-ranking convention: after migration, each destination resorts residents by income, assigns new ranks $r \in (0,1)$, and then sets incomes by $y_r^i = (F_t^i)^{-1}(r)$. Workers ' expectations respect the Calvo process; when evaluating destinations, a household anticipates retaining its current rank r with probability ρ and drawing a fresh rank $r' \sim \text{Unif}(0,1)$ with probability $1-\rho$. They do not internalize the subsequent mechanical renormalization of everyone's ranks that occurs when inflows arrive. This implies that the distribution of non-agricultural productivity remains constant across regions. An alternative specification could track individual types across regions without re-ranking; this would alter the non-agricultural productivity distribution through composition effects and yield sorting-type income distributions, at the cost of additional state variables and loss of closed form.

Agriculture. Each region i is endowed with a fixed supply of land X^i that consists of a continuum of heterogeneous plots ω. Agriculture produces using Cobb-Douglas technology at the plot level:

$$q_t^i(\omega) \;=\; \left[L_t^i(\omega)\right]^{\alpha} \left[z_t^i(\omega)\, X_t^i(\omega)\right]^{1-\alpha},$$

where $L_t^i(\omega)$ and $X_t^i(\omega)$ denote labor and land used on plot ω, and $z_t^i(\omega)$ is plot productivity.

Plots differ in their productivity, and operating a plot requires a per-period cost $z^{i0}(\omega)$, paid in units of the non-agricultural numeraire. Plot heterogeneity is governed by a (nested) Fréchet specification. Let $A_t^i > 0$ and $F^{i0} > 0$ be scale parameters for productivity and operating cost, and let $\theta > 1$ be the common shape parameter. The joint CDF of $(z_t^i(\omega), z^{i0}(\omega))$ is

$$\Pr\Big(z_t^i(\omega) \le z,\; z^{i0}(\omega) \le z^0\Big) \;=\; \exp\left\{ -\, o \left[\left(\frac{z}{A_t^i}\right)^{-\theta} + \left(\frac{z^0}{F^{i0}}\right)^{-\theta} \right] \right\},$$

with the normalization constant $o \equiv \Gamma\big(1+\frac{1}{\theta}\big)^{-\theta}$ so that $A_t^i = \mathbb{E}[z_t^i(\omega)]$ and $F^{i0} = \mathbb{E}[z^{i0}(\omega)]$. Here, A_t^i is the region-specific unconditional mean productivity (fundamental productivity). Because high-productivity plots are more likely to be cultivated, the average productivity of cultivated land (i.e., measured productivity $\tilde{A}_t^i$) exceeds A_t^i unless all plots are cultivated.

Non-agriculture. Non-agriculture consists of a continuum of jobs η that produce a homogeneous good using labor linearly:

$$q_t^{i;n}(\eta) \;=\; A^{i;n}(\eta)\, L_t^{i;n}(\eta),$$

where $A^{i;n}(\eta)$ is job-specific productivity and $L_t^{i;n}(\eta)$ is labor assigned to job η. Regional productivity distributions are i.i.d. log-normal,

$$\ln A^{i;n}(\eta) \sim \mathcal{N}\big(\mu^i,\; (\sigma^i)^2\big),$$

where μ^i and σ^i are region-specific parameters. μ shifts the mean productivity, while σ governs dispersion of productivity across non-agricultural jobs.

5.2.2. Temporary equilibrium

A *temporary equilibrium* in period t is a collection of agricultural prices $\{p_t^{j;a}\}_{j\in\mathcal{S}}$, the non-agricultural price p_t^0, agricultural and non-agricultural wages $\{w_t^{i;a}, w_t^{i;n}\}$, labor allocations $\{L_t^{i;a}, L_t^{i;n}\}$, cultivated land $X_t^{i;a}$, outputs $\{Q_t^{i;a}, Q_t^{i;n}\}$, and regional income distributions $\{f_t^i(y)\}$ such that the following conditions hold:

(i) **Household demand.** Households, endowed with income net of migration costs $\tilde{y}_t^i$, maximize Cobb-Douglas utility, choosing $C_t^{i;a}$ and $C_t^{i;n}$ subject to

$$P_t^{i;a} C_t^{i;a} + p_t^0 C_t^{i;n} \le \tilde{y}_t^i, \tag{5}$$

implying that a fraction γ of income is spend on agriculture and $1-\gamma$ on non-agriculture.

$P_t^{i;a}$ is the CES price index for the agricultural composite in region i:

$$P_t^{i;a} = \left(\sum_{j \in \mathcal{S}} \psi^j \left(p_t^{j;a} \delta^{j,i} \right)^{1-\epsilon} \right)^{1/(1-\epsilon)}, \tag{6}$$

where $\delta^{j,i}$ represents iceberg trade costs.

CES demand implies import expenditure shares

$$\pi_t^{j,i;a} \equiv \frac{\psi^j (p_t^{j;a} \delta^{j,i})^{1-\epsilon}}{\sum_{k \in \mathcal{S}} \psi^k (p_t^{k;a} \delta^{k,i})^{1-\epsilon}}, \tag{7}$$

where $\pi_t^{j,i;a}$ denotes the share of region i's agricultural expenditure allocated to variety j.

(ii) **Labor allocation & wages.** Agricultural wages are equal to the marginal product of labor on the marginal plot:

$$w_t^{i;a} = p_t^{i;a} \alpha \left(\frac{A_t^i \left(\zeta_t^i \right)^{-1/\theta} X_t^{i;a}}{L_t^{i;a}} \right)^{1-\alpha}, \tag{8}$$

where ζ_t^i is the share of land cultivated in region i at time t.

Workers accept a non-agricultural job if and only if $A^{i;n}(\eta) \geq w_t^{i;a}$. The agricultural employment share is hence:

$$\lambda_t^i = \Pr(A^{i;n}(\eta) < w_t^{i;a}) = \Phi\Big(\tfrac{\ln w_t^{i;a} - \mu^i}{\sigma^i} \Big), \tag{9}$$

where Φ is the standard normal CDF. Agricultural and non-agricultural labor are then determined as $L_t^{i;a} = \lambda_t^i L_t^i$ and $L_t^{i;n} = (1 - \lambda_t^i) L_t^i$.

The average non-agricultural wage is determined by the distribution of jobs that are filled:

$$w_t^{i;n} \equiv E\big[y \mid y \geq w_t^{i;a} \big] = \exp\Big(\mu^i + \tfrac{1}{2} (\sigma^i)^2 \Big) \, \frac{1 - \Phi\big(\frac{\ln w_t^{i;a} - \mu^i - (\sigma^i)^2}{\sigma^i} \big)}{1 - \Phi\big(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i} \big)}.$$

This combined wage structure implies that the income distribution in region i, denoted $f_t^i(y)$, follows a truncated log-normal distribution with a mass point at $w_t^{i;a}$.

(iii) **Land use & agricultural output.** Profit maximization implies that plot ω is cultivated if

$$z_t^i(\omega) \Omega_t^i \;\geq\; p_t^0 z^{i0}(\omega), \tag{10}$$

where

$$\Omega_t^i \equiv \Big[\alpha^\alpha (1-\alpha)^{1-\alpha} p_t^{i;a} (w_t^{i;a})^{-\alpha} \Big]^{\frac{1}{1-\alpha}} \tag{11}$$

captures the effective profitability of agriculture in region i at time t.

The share of cultivated land is given by

$$\zeta_t^i = \frac{(A_t^i \Omega_t^i)^\theta}{(A_t^i \Omega_t^i)^\theta + (F^{i0} p_t^0)^\theta}, \tag{12}$$

which follows from the extreme-valued distribution of productivity and operating costs across fields. The share of cultivated land together with the total land area determines agricultural land: $X_t^{i;a} = \zeta_t^i X^i$. Aggregate agricultural output then follows as:

$$Q_t^{i;a} = (L_t^{i;a})^\alpha \left[A_t^i (\zeta_t^i)^{-1/\theta} X_t^{i;a}\right]^{1-\alpha}, \tag{13}$$

where $A_t^i(\zeta_t^i)^{-1/\theta}$ captures that only the most productive fields are used; hence, measured agricultural productivity differs from the fundamental (average) productivity of land in region i at time t.

(iv) **Non-agricultural output.** Only jobs associated with wages above $w_t^{i;a}$ are filled. Aggregate production of the non-agricultural sector in region i at time t is

$$Q_t^{i;n} = w_t^{i;n}(1 - \lambda_t^i) L_t^i, \tag{14}$$

where $w_t^{i;n}$ is the conditional mean wage of accepted jobs.

(v) **Goods market clearing.** Agricultural and non-agricultural markets clear:

$$p_t^{j;a} Q_t^{j;a} = \gamma \sum_{i \in \mathcal{S}} \pi_t^{j,i;a} Y_t^i, \tag{15}$$

$$\sum_{j \in \mathcal{S}} p_t^0 Q_t^{j;n} = (1-\gamma) \sum_{i \in \mathcal{S}} Y_t^i, \tag{16}$$

where $\pi_t^{j,i;a}$ are CES import shares, and Y_t^i denotes regional income. I assume that the total income in region i at time t equals the wage bill plus agricultural profits, land-clearing costs, and migration costs, and that all income is spent locally on consumption goods.

Given labor allocations across regions $\{L_t^i\}$, the temporary equilibrium determines wages, land use, and income distributions, which feed back into migration decisions in the dynamic block.

6. Estimation and calibration

6.1. Calibration of the baseline economy

The baseline economy is calibrated to 2010 under the assumption that the static trade model is in temporary equilibrium. Several parameters are sourced from the literature, while the remaining ones are disciplined by matching model-implied moments with observed data.

Exogenous parameters. The Armington elasticity is set to $\epsilon = 9$ (Allen and Arkolakis, 2014); and the dispersion of land productivity to $\theta = 2$, the midpoint of available estimates (Sotelo, 2020; Costinot, Donaldson, and Smith, 2016). Interregional iceberg trade costs are parameterized as $\delta^{j,i} = (1 + \text{distance}^{j,i})^{0.08}$, implying around 65% higher prices at 500 kilometers from the producing region (Pellegrina, 2022).

Inversion from data. Observed inputs include the land endowments X_0^i, cultivated share of land ζ_0^i, regional labor endowments L_0^i and sectoral allocations $L_0^{i;a}, L_0^{i;n}$, sectoral wages $w_0^{i;a}, w_0^{i;n}$, measured agricultural productivity among cultivated plots $(A_t^i \left(\zeta_t^i\right)^{-1/\theta})$, and agricultural value of production $p_0^{i;a} Q_0^{i;a}$.

From these, I back out: the Cobb-Douglas share α from the agricultural wage bill-to-output ratio (0.1579); the distributional parameters (μ^i, σ^i) of non-agricultural productivity distributions are obtained by matching the agricultural labor share λ_0^i and the observed non-agricultural mean wage; the fundamental land productivity A_0^i is obtained by adjusting measured productivity for endogenous selection; producer prices $p_0^{i;a}$ are recovered by inverting the agricultural wage equation; and agricultural fixed costs F^{i0} are derived from the optimal land-use condition.

Residual calibration. Given these objects, agricultural and non-agricultural outputs $(Q_0^{i;a}, Q_0^{i;n})$, land-clearing costs B_0^i, agricultural profits R_0^i, and total regional income Y_0^i are recovered. The agricultural expenditure share γ is set to match the ratio of agricultural spending to total income (0.1190). Finally, region-specific preference shifters ψ^j are solved as a fixed point problem to clear the agricultural goods market and are interpreted as quality or taste parameters.

Table II summarizes all calibrated objects, their interpretation, and their empirical targets. Appendix E.1 presents additional information on the calibration and model fit.

6.2. Counterfactual fundamental productivity

To simulate climate change, I translate projected temperatures into counterfactual changes in fundamental agricultural productivity. I recover baseline productivity A_0^i during the calibration process (Section 6.1). I then obtain counterfactual values A_t^i by scaling these baselines using the estimated quadratic temperature-productivity relationship shown in Figure IV.

Let $\widehat{g}(TG) = \hat{\beta}_1 TG + \hat{\beta}_2 TG^2$ represent the fitted quadratic response of $\ln A_{it}$ to temperature. For each region i and year t, I compute

$$\ln A_t^i = \ln A_0^i + \left(\widehat{g}(TG_{it}) - \widehat{g}(TG_{i0})\right),$$

where TG_{i0} is the region's baseline temperature in 2010.

This procedure preserves cross-sectional heterogeneity in A_0^i and allows productivity dynamics to reflect the nonlinear temperature response. Since each region is anchored to its own baseline temperature, cooler and hotter regions move along different segments of the quadratic curve. This approach implies heterogeneous productivity impacts from a common warming

TABLE II
Calibration of the baseline economy, 2010

Object	Interpretation	Target / Source	Value
ϵ, θ	Trade elasticities, heterogeneity in land quality	Literature	9, 2
α	Labor share in agriculture	Agricultural wage bill / value of production	0.1579
γ	Agricultural expenditure share	Agricultural spending / income	0.119
Varying by region:			
μ^i	Location parameter (non-agricultural productivity)	$w_0^{i;a}, w_0^{i;n}, \lambda_0^i$	9.36 (0.43)
σ^i	Scale parameter (non-agricultural productivity)	$w_0^{i;a}, w_0^{i;n}, \lambda_0^i$	0.64 (0.18)
A_0^i	Fundamental land productivity	Corrected measured productivity	660.12 (524.03)
$p_0^{i;a}$	Agricultural producer price	Inverted from wage equation	17.08 (21.34)
F^{i0}	Fixed cost of land clearing	Land share condition	7109.07 (22908.90)
ψ^j	Regional preference shifter	Agricultural goods market clearing	0.01 (0.08)
$\delta^{j,i}$	Trade costs	Distance data	1.78 (0.12)

Notes. The table summarizes the calibration of the baseline economy for 2010. Exogenous parameters are taken from the literature or distance data. Endogenous parameters and objects are identified using the static model's temporary equilibrium conditions and matched to Brazilian census and agricultural statistics. For regionally varying parameters, the first value reported is the mean across all regions, and the value in parentheses is the standard deviation.

shock. I use projected temperatures TG_{it} under SSP5-8.5 to generate the counterfactual productivity paths for the simulations.

In the model, I measure five-year periods from 2010 to 2100, with $t = 0$ corresponding to 2010. I average temperature projections over five-year windows to obtain A_t^i for $t = 0, 1, \ldots, 18$. My estimates indicate that, by 2100, some regions will lose up to 55 percent of initial productivity, while others, particularly those with low starting temperatures, will gain up to 38.5 percent. Table III presents the projected distribution of productivity changes across mesoregions for selected years.

TABLE III
Projected deviation from baseline productivity by year

Year	Median change	Min change	Max change
2030	-1.5%	-8.3%	13.1%
2050	-4.9%	-16.4%	20.8%
2100	-24.2%	-55.3%	38.5%

Notes. This table reports the median, minimum, and maximum change in mesoregions' fundamental agricultural productivity for 2030, 2050, and 2100 (relative to 2010). Counterfactual productivity levels are obtained by applying the estimated quadratic temperature-productivity relationship (from Figure IV) to projected annual mean temperatures under the SSP5-8.5 scenario, simulated with the CMCC-ESM2 Earth System Model from the Copernicus Climate Change Service (C3S) CMIP6 archive. Baseline fundamental productivity for 2010 is recovered in the calibration of the model. Projections use five-year averaged temperatures expressed as anomalies relative to 2010.

The full path of projected deviations from the baseline is shown in Appendix E.2.

6.3. Dynamic part & migration costs

The dynamic component of the model requires specification of the following parameters: β (discount rate), ν (variance of idiosyncratic location preferences), ρ (persistence of non-agricultural productivity draws), $\{\chi^{j,i}\}_{j,i\in\mathcal{S}}$ (bilateral matrix of monetary migration costs), and $\{\kappa^j\}_{j\in\mathcal{S}}$ (origin-specific non-monetary cost of leaving home).

Exogenous parameters. Following Cai, Caliendo, Parro, and Xiang (2022), the discount factor β is set to 0.86, which corresponds to an annual discount factor of 0.97. Rank persistence is fixed at $\rho = 0.9$, consistent with the low intergenerational mobility observed in Brazil (Britto, Fonseca, Pinotti, Sampaio, and Warwar, 2022). The idiosyncratic taste scale is set to $\nu = 1.75$ (Caliendo et al. (2019) find an annual elasticity of 2.02 and discuss that this value must be larger at higher frequencies).[11]

Recovering migration costs. To recover migration costs, I match model-implied aggregate migration probabilities between regions in the first period to the observed migration probabilities in the data. For this, I solve the full dynamic model under alternative cost shifters and select the cost shifter that best aligns the model-implied probabilities with its data counterpart.[12]

The model is implemented as a finite-horizon problem, beginning in 2010 and ending in 2100. Each period represents five years, resulting in a total of 18 periods. The state variable is defined as the rank in the non-agricultural productivity distribution r. For tractability, each region's non-agricultural productivity distribution is discretized into 100 equiprobable bins, with each bin's average implied income (y_r) used as the representative state for computing continuation values and migration feasibility/costs. To calibrate costs, I assume that workers expect agricultural productivity to evolve according to the pessimistic warming scenario SSP5-8.5.

Migration costs are assumed to consist of a variable distance component and an origin-specific fixed cost:

$$\chi^{j,i} = \theta^{\text{dist}} \cdot \text{distance}^{j,i},$$
$$\tau^{j,i}(y) = \begin{cases} -\log(1 - \chi^{j,i}/y) + \kappa^j \cdot \mathbf{1}\{i \neq j\}, & y > \chi^{j,i}, \\ \infty, & y \leq \chi^{j,i}, \end{cases}$$

[11] Typically, values of ν are inferred from regressions of migration flows on wage differentials. A weak empirical response of migration to wage gaps is interpreted as evidence of a high ν, implying that idiosyncratic taste dispersion $1/\nu$ is low. However, in my model, low migration responsiveness arises not from strong idiosyncratic preferences but from mobility constraints. Under this interpretation, existing empirical approaches would overestimate ν, attributing limited migration responses to high preference heterogeneity rather than to constraints. This reasoning is consistent with the observation that studies such as Cruz (2024) tend to find higher values of ν for low-income countries. It seems unlikely that individuals in these contexts have stronger idiosyncratic preferences; a more plausible explanation is that they face greater barriers to responding to wage differentials. A value of 1.75, hence, seems to be in the plausible range of values once accounting for this measurement problem.

[12] I assume that migration probabilities between 2005 and 2010 recovered from the 2010 Census are representative of probabilities from 2010 to 2015.

Here, θ^{dist} determines the rate at which monetary costs increase with distance, and κ^j captures non-monetary "home" preferences at origin j. By construction, migration is disproportionately costly for individuals with low incomes: $\tau^{j,i}(y)$ approaches infinity as y approaches $\chi^{j,i}$ from above, and migration becomes infeasible if income does not cover the monetary cost.

Solution algorithm. Conditional on β, ρ, and ν, the parameters θ^{dist} and κ^j are estimated to jointly determine migration costs under the parametric assumptions described above. Formally, the minimization problem is:

$$\min_{\theta^{\text{dist}}, \{\kappa^j\}_{j \in \mathcal{S}}} \sum_{j,i} \left(\mu_0^{j,i}(\theta^{\text{dist}}, \kappa^j) - \mu_{0,\text{data}}^{j,i} \right)^2,$$

where $\mu_{0,\text{data}}$ are observed bilateral migration probabilities.

The calibration process is computationally intensive, as it requires repeatedly solving the model for different cost values. To maintain feasibility, the search over θ^{dist} is conducted using a grid search from R\$10/km to R\$60/km in increments of R\$10/km. Conditional on each value of θ^{dist}, the non-monetary costs of leaving home, κ^j, are estimated non-linearly to match each region's overall emigration rate.

Results. A value of θ^{dist} equal to R\$50/km provides the best fit to bilateral migration probabilities. The fixed costs κ^j have a mean value of 6.69, indicating a high psychic cost of leaving home, equivalent to approximately 96 percent of annual income. The R-squared from regressing model-implied migration probabilities on observed probabilities exceeds 0.99, with a slope coefficient close to one. When excluding the probability of remaining in the same region and focusing on bilateral migration probabilities to other regions, the R-squared is 0.43 with a slope of 0.77. Appendix E.3 Figure E.6 presents the associated model fit.

Untargeted moments. The calibration targets bilateral migration probabilities rather than aggregate flows. In Table IV, model-implied and observed flows are compared after aggregation into three categories: within-mesoregion, between-mesoregion but within-state, and between-states.

TABLE IV
Fit of migration flows

Category	Model	Data
Same Mesoregion	95.5%	95.6%
Different Mesoregion, Same State	1.8%	1.8%
Different State	2.7%	2.6%

Notes. This table compares the fit of the model-implied and observed migration flows between Brazilian mesoregions. Observed flows represent the mass of workers moving between 2005 and 2010, constructed from the 2010 Population Census based on individuals' reported municipalities of residence in 2005 and 2010, aggregated into mesoregion-to-mesoregion transitions. Model-implied flows correspond to the mass of movers generated in the first period of the calibrated dynamic spatial model, which incorporates the estimated monetary and non-monetary migration costs. Monetary costs increase linearly with geographic distance (R\$50 per km), and origin-specific non-monetary "home" preferences are calibrated to match regional emigration rates.

The model closely matches the observed distribution of flows. Overall, the calibration demonstrates that the model can replicate key features of observed migration patterns. Appendix E.3 shows further statistics of the model fit and a related discussion of the calibration.

7. Welfare losses from climate change

This section quantifies how temperature-induced productivity shocks translate into welfare losses across mesoregions and income groups. The analysis addresses three primary questions: the spatial concentration of climate-induced welfare losses, the households most affected, and the role of migration in mitigating these losses. I first document the spatial and distributional incidence of climate-induced welfare losses before turning to the role of migration as a potential adaptation channel.

Measurement and counterfactuals. The welfare effects of climate change are measured by contrasting expected lifetime utility between a scenario in which regional productivities remain at 2010 levels and one in which they evolve under projected warming. In the latter scenario, the temperature-productivity mapping from Section 6 implies that already-hot regions become less productive, while cooler regions experience modest gains as they approach the agronomic optimum.

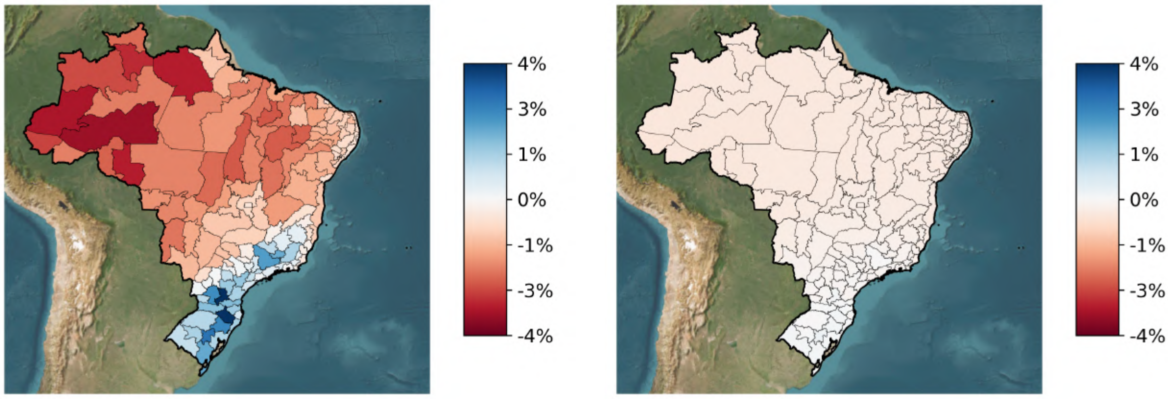
Welfare differences are expressed in consumption-equivalent terms, defined as the constant proportional and permanent increase in consumption required to make a household under climate change as well off as in the no-climate-change counterfactual. This metric represents the uniform percentage change in consumption applied in every period from 2010 to 2100 that equalizes lifetime utility across the two scenarios.

7.1. Quantifying welfare losses

Figure VI presents welfare changes by income percentile across space. Panel (a) shows outcomes for households in the lowest income percentile, and panel (b) for those in the top. For households at the bottom of the income distribution, spatial heterogeneity is pronounced. In the North and Northeast, where rising temperatures sharply reduce agricultural productivity, welfare declines by up to 4%. Conversely, some areas in the South experience welfare gains of similar magnitude as moderate warming increases productivity toward the agronomic optimum. These gains result from favorable local shocks rather than improvements in aggregate welfare.

Among households at the top of the income distribution, welfare effects are negligible across all regions. High-income households are primarily employed in high-productivity non-agricultural jobs and are largely insulated from changes in local agricultural wages and the expansion of low-productivity non-agricultural employment. Their modest welfare losses arise mainly from price effects, as climate change increases the cost of the consumption bundle. Overall, the figure demonstrates a steep distributional gradient and significant within-region heterogeneity; in many cases, welfare differences within a region are as large as those observed across regions.

FIGURE VI
Welfare changes by income percentile



(A) Lowest income percentile (B) Highest income percentile

Notes. Each panel maps the consumption-equivalent welfare change in 2010 relative to a no-climate-change baseline. Panel (a) reports results for the lowest income percentile in each region, and panel (b) reports results for the highest. Welfare changes are expressed as permanent percentage shifts in consumption that equalize lifetime utility across scenarios. Negative values indicate welfare losses; positive values indicate gains.

Table V summarizes welfare changes by major region and income group in 2010. To facilitate comparison across spatial and income dimensions, the 100 income percentiles within each region are categorized into three groups: R$0-10k, R$10-20k, and above R$20k, corresponding to low-, middle-, and high-income households, respectively. In 2010, 51% of the sample of workers fell into the lowest income group, 26% in the middle-income group, and 22% in the highest income group.

The top panel of Table V presents population-weighted mean welfare changes by region and income group. Across all regions, the largest effects are observed among poorer households, indicating that climate damages in hot regions are strongly regressive. In the North, households earning below R$10,000 per year experience nearly 0.9% losses in consumption-equivalent welfare, compared to only 0.3-0.4% among high-income households. Similar gradients are evident in the Northeast (-0.6% versus -0.2%) and Center-West (-0.7% versus -0.2%). These disparities arise because poorer households, while adjusting out of agriculture, are absorbed into low-return local employment. Declining agricultural wages thus trigger a reallocation that mitigates but does not reverse income losses, resulting in persistently larger welfare declines among the poor.

Regional averages, as reported in the top panel, demonstrate substantial spatial variation. Mean welfare losses are greatest in the tropical North and Northeast, at -0.55% and -0.40%, respectively. The Center-West experiences smaller average losses (-0.24%). In contrast, the temperate South records modest welfare gains of $+0.24\%$, as moderate warming increases local productivity toward the agronomic optimum and raises agricultural wages. The Southeast exhibits near-zero mean effects (-0.02%), reflecting the limited dependence of incomes on local agricultural productivity in these predominantly urban regions and the weak transmission of climate shocks through local labor markets.

The bottom panel of Table V reveals the extent of within-region heterogeneity that regional

TABLE V
Welfare losses by region and income bin

	R$ 0–10k	R$ 10–20k	R$ >20k	All households
Population-weighted means				
Norte	-0.87	-0.36	-0.34	-0.55
Nordeste	-0.57	-0.24	-0.23	-0.40
Centro-oeste	-0.68	-0.21	-0.17	-0.24
Sudeste	+0.09	-0.03	-0.03	-0.02
Sul	+0.77	+0.15	+0.13	+0.24
Maximum losses				
Norte	-3.23	-0.90	-0.90	-3.23
Nordeste	-2.34	-0.67	-0.45	-2.34
Centro-oeste	-2.13	-0.47	-0.43	-2.13
Sudeste	-1.62	-0.35	-0.30	-1.62
Sul	-0.63	-0.11	-0.12	-0.63

Notes. Entries report consumption-equivalent welfare changes (percent) in 2010 relative to a no-climate-change baseline, by major region and income bin. Income bins aggregate 100 model percentiles per region into three observed 2010 income groups. The top panel reports population-weighted mean welfare changes; the bottom panel reports the maximum welfare loss in each region-income cell. Negative values indicate welfare losses, and positive values indicate welfare gains. The final column reports the mean or maximum across all households in each region.

averages conceal. It reports the largest welfare loss experienced by any income group within each region. In the North, the most adversely affected group loses over 3% in consumption-equivalent welfare, while the corresponding figures for the Northeast and Center-West are approximately 2-2.5%. Even in the Southeast, where mean effects are nearly zero, the most exposed groups experience declines of about 1.6%. The South remains largely protected from significant losses, attributable to its cooler baseline temperatures, higher incomes, and more diversified economic structure.

Collectively, these results indicate a clear pattern: climate change imposes the greatest welfare costs on poor households in already hot regions, while wealthier and more urbanized areas are largely insulated. The interplay of spatial exposure, sectoral composition, and income inequality produces a double asymmetry across both geography and the income distribution, shaping the aggregate welfare burden of warming.

7.2. Migration as adaptation?

The preceding results show that welfare losses are concentrated among lower-income households in hotter regions. This subsection evaluates migration as a potential adaptation strategy. Table VI presents how migration opportunities differ across the income distribution under climate change. For each year t, the analysis calculates (i) the proportion of households whose income is insufficient to meet the minimum cost of migrating to another region and (ii) the average probability of out-migration. The reported values are population-weighted averages over the entire simulation horizon (2010-2100). The first column reflects the prevalence of liquidity constraints that inhibit relocation, while the second column presents the corresponding realized migration rates under climate change.

TABLE VI
Migration constraints and mobility rates under climate change

Income group	Share unable to migrate ($\chi \geq y$)	Mean migration probability
Low income ($\leq$10k)	52.43%	5.41%
Middle income (10-20k)	6.48%	6.39%
High income (>20k)	0.38%	11.77%

Notes. Each entry reports the population-weighted average across time periods under the climate scenario. Households are classified as *unable to migrate* when their income $y_{i,t}$ falls below the minimum off-diagonal migration cost $\min_{j \neq i} \chi^{ij}$ from their origin. Migration probabilities are defined using off-diagonal elements of the migration matrix μ_t^{ij}.

The table shows that migration opportunities are sharply stratified by income. Roughly half of low-income households are liquidity-constrained, compared with fewer than 7% of middle-income and virtually none of high-income households. Mobility rates mirror these gaps: only about 5% of low-income households migrate in an average year, versus nearly 12% for the high-income group. Hence, migration plays only a minor role as an adaptation mechanism—not because climate shocks fail to generate incentives to move, but because a large share of poor households is unable to finance relocation. The constraint operates primarily on the extensive margin: many low-income households are excluded from migration altogether, rather than marginally adjusting their mobility rates in response to climate shocks.

7.3. Discussion

The pattern of welfare changes reflects the interaction of two central adjustment margins in the model. (i) *Production margin:* Climate change reduces agricultural productivity in hot regions, which lowers reservation wages and shifts marginal workers into low-productivity non-agriculture. This process compresses mean income and increases income dispersion within affected regions, resulting in regressive welfare losses. (ii) *Migration margin:* Because migration requires a monetary cost to be paid, many low-income households are liquidity-constrained even before the shock. Climate change intensifies incentives to relocate but does not alleviate these preexisting constraints; thus, migration contributes little as an adaptation channel.

When liquidity constraints are binding, adaptation through migration becomes inherently limited. Households with the greatest incentives to relocate are often the least able to do so. Therefore, the distributional impact of climate change is shaped as much by monetary frictions as by geography.

8. Policy Experiment: Income-Targeted Migration Subsidy

Climate change disproportionately harms low-income households in hot regions—the very households facing the steepest migration barriers. To evaluate an adaptation policy addressing this challenge, a counterfactual reform is simulated that subsidizes the *monetary* cost of migration for low-income households while leaving non-monetary barriers unchanged. The reform targets liquidity constraints by equalizing the utility-equivalent migration cost for a low-income household with that of a higher-income reference group. As a result, the policy reduces income-based disparities in mobility and enables migration as an adaptation strategy to climate change.

Policy mapping and implementation. Let $\tau^{j,i}(y;\chi^{j,i})$ denote the utility-equivalent cost of moving from origin j to destination i for a household with income y, facing a monetary cost $\chi^{j,i}$ and a non-monetary cost κ^j embedded in $\tau^{j,i}(\cdot)$. Equalizing utility-equivalent costs between a low-income group y_r and a higher-income reference y_h requires a subsidy $\xi^{j,i}$ that satisfies the following condition:

$$\tau^{j,i}(y_r;\chi^{j,i}-\xi^{j,i}) = \tau^{j,i}(y_h;\chi^{j,i}), \tag{17}$$

which, under log utility, implies

$$\xi^{j,i}(y_r \mid y_h) = \chi^{j,i}\Big(1-\frac{y_r}{y_h}\Big), \qquad \text{for } y_r < y_h, \tag{18}$$

and $\xi^{j,i} = 0$ otherwise. Intuitively, the subsidy scales the out-of-pocket cost in proportion to the income gap relative to the reference type.

In the baseline experiment, the reference income is set at $y_h = \text{R\$}\,20{,}000$. Only households with $y_r < y_h$ are eligible for the subsidy, and higher-income households are excluded. The subsidy is provided in every period from 2010 to 2100.

8.1. Welfare gains from subsidy

Table VII presents consumption-equivalent welfare changes by income group. Under projected climate change, reducing monetary migration costs yields significant welfare gains for low-income households and comparatively smaller gains for higher-income groups. Lifetime welfare rises by 24.09% for households with incomes below R$10k, 5.49% for those with R$10–20k, and 1.33% for those above R$20k. The welfare gains for households above R$20k, who do not receive the subsidy, arise from two mechanisms. First, the migration subsidy facilitates a more efficient allocation of labor across regions and lowers the price of the consumption bundle, which increases real income for high-income households. Second, although these households are initially wealthy, they remain exposed to the risk of having a low non-agricultural productivity draw in the future. As a result, lifetime welfare incorporates the insurance value the policy provides against potential future income losses.

The second row shows the proportion of baseline climate losses mitigated by the policy. Here, the climate losses mitigated by the policy represent the difference-in-differences in welfare losses from climate change with and without the subsidy. A positive value indicates that welfare losses from climate change are lower with the subsidy than under the baseline mobility scenario. The subsidy offsets approximately 4.2% of baseline losses for the two lowest-income groups and 1.0% for the highest-income group.

TABLE VII
Welfare gains and attenuation of climate losses from a migration subsidy

	R$ 0–10k	R$ 10–20k	R$ >20k	All households
Subsidized vs. Baseline (CEV, %)	24.09	5.49	1.33	8.22
Share of climate losses removed (%)	4.16	4.10	1.01	2.92

Notes. Entries report population-weighted mean consumption-equivalent welfare changes (percent) under climate change. The first row compares welfare between the subsidized and baseline migration-cost regimes, holding the climate path fixed. The second row reports the share of baseline climate losses offset by the policy, defined as the reduction in welfare losses relative to baseline losses. The last column reports the national average across all households.

Table VIII presents welfare gains disaggregated by macro-region and income group. The Northeast demonstrates the largest gains, with low-income households experiencing welfare increases exceeding 36 percent. This result is driven by two reinforcing factors: high climate exposure in agriculture and significant baseline migration barriers that limit out-migration without the subsidy. The North and Center-West also show substantial low-income gains, at 14 percent and 9 percent, respectively. The South and Southeast display similar gains, with both regions at approximately 11 percent.

TABLE VIII
Welfare gains by region

	R$ 0–10k	R$ 10–20k	R$ >20k	All households
Climate change in both scenarios				
Subsidized vs. Baseline (CEV, %)				
Norte	14.11	6.05	2.94	8.30
Nordeste	36.41	13.70	4.94	23.44
Centro-oeste	9.37	4.87	1.29	3.68
Sudeste	11.40	2.66	0.56	2.68
Sul	11.17	4.33	1.09	4.19

Notes. Entries report consumption-equivalent welfare gains (percent) from the migration subsidy under climate change, by macro-region and income bin. Each entry compares welfare under the subsidized and baseline regimes, holding the climate path fixed. The last column reports the regional population-weighted mean. "Income bins" correspond to 2010 income groups used throughout the paper (R$0-10k, R$10-20k, and above R$20k).

Table IX provides evidence on the mechanisms underlying these welfare gains by reporting how the subsidy affects migration feasibility and mobility rates. The first column presents the proportion of households with incomes below the minimum monetary migration cost $\min_{j \neq i} \chi^{ij}$, and the second column displays average migration probabilities. Under the baseline calibration, approximately half of low-income households face binding liquidity constraints that prevent them from migrating. The introduction of the subsidy substantially relaxes these constraints: the proportion unable to migrate declines from 52% to below 4%, and the average migration probability nearly doubles. Middle-income households experience a smaller yet significant increase in mobility, whereas high-income households remain largely unaffected because they were not initially constrained.

TABLE IX
Effect of the migration subsidy on migration constraints and mobility (climate path)

Income group	Share unable to migrate ($\chi \geq y$)	Mean migration probability
Low income ($\leq$10k)	52.43% $\rightarrow$ 3.90%	5.41 $\rightarrow$ 11.09%
Middle income (10-20k)	6.48% $\rightarrow$ 2.56%	6.39 $\rightarrow$ 5.41%
High income (>20k)	0.38% $\rightarrow$ 0.38%	11.77 $\rightarrow$ 9.63%

Notes. Each entry compares the baseline and subsidized migration-cost regimes under the same climate path. Under the subsidy, all households with baseline income below R$20,000 are treated as if they had income equal to R$20,000 when facing monetary migration costs χ^{ij}. "Share unable to migrate" reports the fraction of households whose income falls below the minimum off-diagonal migration cost $\min_{j \neq i} \chi^{ij}$; "Mean migration probability" reports the average off-diagonal migration probability μ^{ij}. Values are weighted by population and averaged over time. Arrows ($\rightarrow$) indicate the change from the baseline to the subsidized regime.

8.2. Fiscal incidence and dynamic payoffs

Table X summarizes the fiscal incidence of the migration subsidy, averaged across all years in which the policy is active (2010-2100). On average, annual transfers represent approximately 1.3% of GDP per period. The distribution of payments is highly progressive. Households earning less than R$10,000 per year receive an average per-capita transfer of R$913. Those with incomes between R$10,000 and R$20,000 receive R$173, while higher-income households receive no transfer.

This pattern arises endogenously from equilibrium migration decisions under the policy. Since the subsidy is provided exclusively to households that migrate, observed transfers reflect both the identity of movers and the magnitude of the top-up required to offset their mobility costs. Low-income households receive higher transfers for two primary reasons. First, they are more likely to migrate in response to climate shocks when liquidity barriers are reduced. Second, the policy rule stipulates that the subsidy scales with the migration cost wedge, defined as the difference between a household's income and the benchmark income of R$20,000. This policy design ensures that lower-income households, for whom the monetary cost of relocation constitutes a larger share of income, receive proportionally greater support.

TABLE X
Average fiscal incidence of the migration subsidy

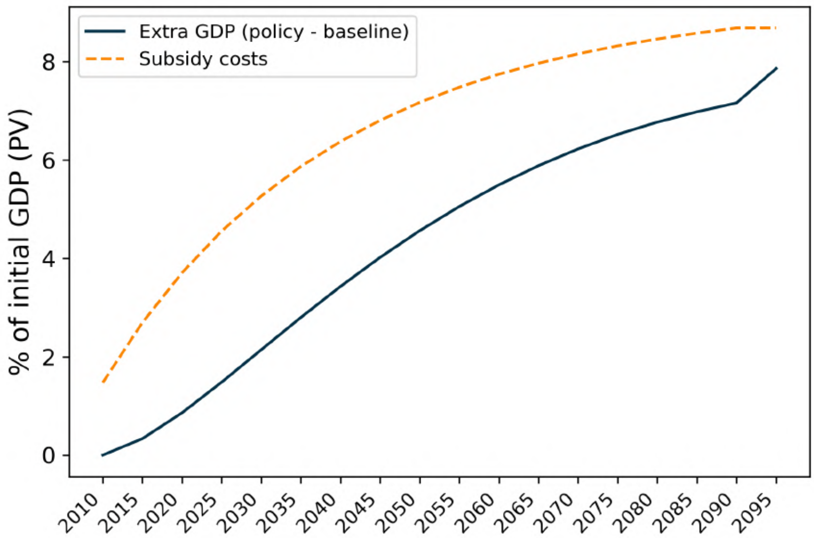
Income bin	Avg. transfer per capita (R$)	Share of GDP (%)
R$ 0–10k	913.3	0.96
R$ 10–20k	173.1	0.32
R$ >20k	0.0	0.00
Total	**271.5**	**1.27**

Notes. Each entry reports the population-weighted average per-capita transfer (R$) and the corresponding fiscal outlay as a share of GDP in the same period, averaged across all years in which the subsidy is active (2010-2100). Fiscal shares are calculated as the mean of annual transfer-to-GDP ratios over the policy horizon. The final row reports the population-weighted national average.

The preceding analysis shows that the migration subsidy generates large welfare gains for low-income households, mitigating a substantial share of their climate-induced losses by enabling relocation to higher-productivity regions. Although these welfare improvements entail a fiscal cost of approximately 1.3% of GDP per period, the policy yields considerable aggregate output gains over time. Figure VII illustrates the evolving relationship between fiscal costs and macroeconomic benefits.

The solid line represents the cumulative present value of additional GDP generated by the subsidy, while the dashed line depicts the cumulative present value of subsidy expenditures, both measured as a share of initial GDP. Over the policy horizon, the program nearly achieves fiscal self-sufficiency. The discounted value of incremental GDP, at 7.9%, offsets approximately 90% of the present value of total transfers, which is 8.7%. The shortfall primarily results from the gradual accumulation of migration-induced output gains as workers relocate to more productive regions, whereas fiscal costs are incurred predominantly at the beginning of the program.

FIGURE VII
Self-financing profile of the migration subsidy



Notes. The solid line shows the cumulative present value of additional GDP generated by the subsidy, and the dashed line shows the cumulative present value of subsidy expenditures. Both are expressed as percentages of initial baseline GDP. Discounting uses a five-year factor of $\beta = 0.86$ (approximately a 3% annual rate). Over 2010-2100, the present value of additional GDP covers about 90% of total transfers, implying an internal rate of return of roughly 13% per model period (2.6% annually).

Aggregate output gains arise from higher output per worker, shown in Table XI. Under baseline migration costs, climate change sharply reduces agricultural productivity: by 2095, agricultural output per worker falls by 12.5%, while non-agricultural output per worker declines only marginally. When the subsidy is introduced, output per worker rises persistently in both sectors. By 2095, output per worker is 6.4% higher in agriculture and over 10% higher in non-agriculture relative to the baseline, reflecting the gradual reallocation of labor from low-productivity, climate-damaged regions to high-productivity destinations. These compositional effects translate large micro-level welfare gains into aggregate efficiency gains, providing the fiscal base that nearly finances the program.

TABLE XI
Output per worker by sector and policy regime (percent differences)

	Year	Agriculture	Non-agriculture	All sectors
Baseline mobility				
Climate change − No climate change	2030	0.71	-0.05	0.04
Climate change − No climate change	2050	-0.59	-0.10	-0.16
Climate change − No climate change	2095	-12.51	-0.40	-1.84
Policy effect under climate change				
Subsidy − Baseline	2030	2.74	1.03	1.23
Subsidy − Baseline	2050	4.30	1.55	1.88
Subsidy − Baseline	2095	6.36	10.48	9.99

Notes. Entries show percent differences in output per worker between scenarios. For each year, "Climate change − No climate change" compares the climate path to the no-climate-change baseline within the same migration regime. "Policy effect under climate" reports the difference between the subsidized and baseline regimes under climate change. The final column aggregates sectors using fixed 2010 GDP weights. Output per worker is defined as sectoral output divided by sectoral employment.

9. Policy implications and broader relevance

Income-targeted migration assistance serves both as a redistributive mechanism and as an adaptation strategy to climate change. Although the policy does not directly counteract the productivity decline associated with rising temperatures, it substantially reduces welfare losses by enabling liquidity-constrained households to relocate from severely affected, low-productivity regions to higher-productivity destinations. Because the most severe climate impacts disproportionately affect poorer households in hotter areas, lowering financial barriers to migration advances both equity and efficiency objectives: it improves the welfare of vulnerable populations while enabling a more efficient allocation of labor in space.

In practice, such a policy could be implemented through existing fiscal and social-protection systems. One option is to provide targeted tax credits to low-income workers who move to regions or sectors with lower climate exposure. While administratively straightforward, this approach would likely exclude informal workers and subsistence farmers who fall outside the formal tax system. Alternatively, migration assistance could be distributed through established social-security programs that already reach low-income households, for example by offering relocation supplements or mobility grants within broader social-protection frameworks.

This framework offers several promising directions for future research. A natural next step is to explicitly incorporate taxation, enabling a unified assessment of how migration subsidies and income transfers interact with the broader fiscal system and how an optimal dynamic redistribution scheme might evolve over time. Such a setting introduces an intertemporal trade-off: early intervention, when climate damages are limited, entails higher initial fiscal costs and yields modest short-term gains, whereas delayed intervention may forgo the adaptive benefits of gradual relocation. The framework could also be used to compare mobility-based interventions with alternative policies such as place-based transfers, infrastructure investments, or benefits linked to residential location. While such programs may improve local welfare in the short run, they can also confine households to high-risk regions. The key policy trade-off is that mobility-promoting measures allow households to avoid exposure by relocating, whereas place-based interventions enhance short-term welfare but may increase long-term vulnerability.

10. Conclusion

This paper demonstrates that income significantly influences households' capacity to respond to climate change through migration. Using Brazilian Census microdata, I documented that within the same origin, higher-income individuals are substantially more likely to migrate. A dynamic spatial general equilibrium model with income-dependent migration costs is developed to interpret these findings and to quantify the distributional welfare consequences of climate change.

The analysis yields three main findings. First, climate change results in strongly regressive welfare impacts. In already-hot regions, low-income households experience consumption-equivalent welfare losses of up to 3.3 percent per period, while richer households in the same regions remain largely unaffected. In contrast, in cooler regions, climate change can increase agricultural productivity, allowing some low-income households to realize modest gains. Second, many low-income households face liquidity constraints that prevent them from financing migration, limiting the role of migration as an adaptation strategy. Third, an income-targeted migration subsidy that reduces the monetary costs of moving for low-income households increases welfare, partially mitigates climate damages, and recoups a substantial portion of expenditures over time through higher output per worker.

Collectively, these results indicate that adaptation to climate change is determined by both geographic and socioeconomic factors. Low-income households are most vulnerable to climate change damages but are least able to use migration as an adaptation strategy. Policies that alleviate liquidity constraints on mobility can serve as both redistributive and climate adaptation measures, complementing place-based interventions that address local productivity shocks. More broadly, the findings imply that, when monetary migration costs and liquidity constraints are present, the distributional effects of economic shocks depend as much on the capacity to respond as on the location of the shocks.

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Online Appendix

Appendices

A. Data

A.1. Census data

The primary data source is the 2010 Brazilian Census, which provides detailed information on individuals' demographics, income, occupation, and migration history. The Census is conducted by the Brazilian Institute of Geography and Statistics (IBGE) and is representative at the mesoregion level.

To retrieve the data, I use IPUMS International. While IPUMS provides harmonized variables, I use the original Census occupation codes to map individuals to agricultural and non-agricultural sectors. IPUMS further allows me to directly retrieve an individual's current mesoregion of residence. The place of residence in 2005 is reported as the municipality of residence in 2005, which I map to mesoregions using the official IBGE correspondence. Income is measured as the total monthly income from all sources, reported in Brazilian Reais (R$) in 2010 (variable `INCTOT`). I annualize income by multiplying the reported monthly income by 12.

Sample selection. The sample consists of Brazilian-born individuals aged 25–64 with positive income and non-missing information on occupation, place of residence in 2005, race, education, marital status, and gender. This yields approximately 6.2 million microdata observations, which correspond to about 60.3 million individuals in the weighted population.

Mapping sectoral employment. I map Census occupation codes to those working in crop cultivation and non-crop related occupations. Throughout the paper, I refer to the former as "agriculture" and the latter as "non-agriculture". This concept of agriculture is narrower than the agricultural sector in national accounts, which includes activities such as livestock, forestry, and fishing. However, to align the model with my measure of agricultural productivity (based solely on crop production), I focus on crop cultivation. The mapping is based on the 2002 Brazilian Classification of Occupations (CBO) and is detailed in Table A.1.

Summary statistics. Table A.2 shows summary statistics for the sample used throughout the paper. The average age is 40 years with 43% female. The majority of individuals are married or in a union (70%) and identify as White (51%) or Brown (39%). Educational attainment is relatively low, with 30% having less than primary education completed and only 15% having completed university. The average annual income is R$ 19,334, with a standard deviation of R$ 59,143. Approximately 8.2% of the sample works in crop cultivation.

TABLE A.1
Mapping of Census occupation codes to agriculture

Code	Occupation
01101	Cultivation of rice
01102	Cultivation of corn
01103	Cultivation of other cereals for grains
01104	Cultivation of herbaceous cotton
01105	Cultivation of sugar cane
01106	Cultivation of tobacco
01107	Cultivation of soybeans
01108	Cultivation of manioc root
01109	Cultivation of other temporary agricultural farm products
01110	Cultivation of salad greens, vegetables and other such plant products
01111	Cultivation of flowers, ornamental plants and greenhouse products
01112	Cultivation of citric fruits
01113	Cultivation of coffee
01114	Cultivation of cocoa beans
01115	Cultivation of grapes
01116	Cultivation of bananas
01117	Cultivation of other permanent farming products
01118	Poorly specified farm crops
01119	Non-specified crop

Notes. The table shows the mapping of Census occupation codes to agriculture. The occupation codes are based on the 2002 Brazilian Classification of Occupations (CBO).

TABLE A.2
Summary statistics

Characteristic	**N = 60,310,390**
Income	Mean: 19,334 (Std.dev.: 59,143)
Age	Mean: 40 (Std.dev.: 10)
Crop worker	4,962,668 (8.2%)
Gender	
Female	26,113,936 (43%)
Male	34,196,453 (57%)
Marital status	
Married/in union	42,416,733 (70%)
Separated/divorced/spouse absent	8,376,029 (14%)
Single/never married	8,477,258 (14%)
Widowed	1,040,370 (1.7%)
Race	
Asian	654,227 (1.1%)
Black	5,058,423 (8.4%)
Brown (Brazil)	23,700,411 (39%)
Indigenous	160,917 (0.3%)
White	30,736,412 (51%)
Educational attainment	
Less than primary completed	17,892,207 (30%)
Primary completed	15,444,009 (26%)
Secondary completed	17,842,030 (30%)
University completed	9,132,143 (15%)

Notes. The table shows summary statistics for the sample of Brazilian-born individuals aged 25-64 with positive income and non-missing information on occupation, place of residence in 2005, race, education, marital status, and gender. The sample consists of approximately 6.2 million microdata observations, which correspond to about 60.3 million individuals in the weighted population.

TABLE A.3
Ex-post characteristics of migrants vs. non-migrants

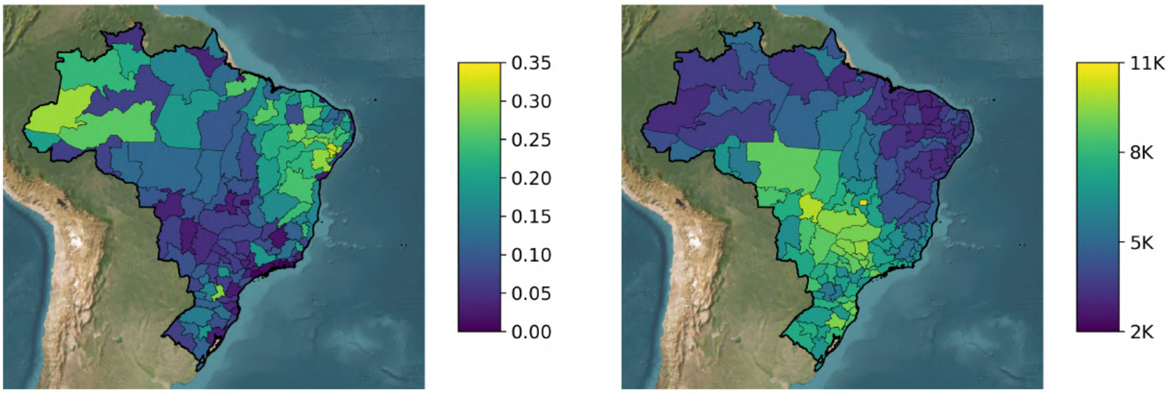
Migrated	**0** N = 57,634,344[1]	**1** N = 2,676,046[1]
Age	Mean: 40 (Std.dev.: 10)	Mean: 36 (Std.dev.: 9)
Income	Mean: 19,046 (Std.dev.: 57,856)	Mean: 25,537 (Std.dev.: 81,863)
Crop worker	4,804,920 (8.3%)	157,749 (5.9%)
Gender		
Female	25,090,645 (44%)	1,023,292 (38%)
Male	32,543,699 (56%)	1,652,754 (62%)
Marital status		
Married/in union	40,573,369 (70%)	1,843,363 (69%)
Separated/divorced/spouse absent	7,941,995 (14%)	434,034 (16%)
Single/never married	8,104,096 (14%)	373,162 (14%)
Widowed	1,014,883 (1.8%)	25,487 (1.0%)
Race		
Asian	621,496 (1.1%)	32,731 (1.2%)
Black	4,857,100 (8.4%)	201,322 (7.5%)
Brown (Brazil)	22,664,755 (39%)	1,035,656 (39%)
Indigenous	153,362 (0.3%)	7,555 (0.3%)
White	29,337,631 (51%)	1,398,782 (52%)
Educational attainment		
Less than primary completed	17,248,562 (30%)	643,645 (24%)
Primary completed	14,770,428 (26%)	673,581 (25%)
Secondary completed	17,058,590 (30%)	783,440 (29%)
University completed	8,556,763 (15%)	575,380 (22%)

Notes. The table shows summary statistics for migrants (1) and non-migrants (0) between 2005 and 2010. Importantly, the characteristics are measured ex-post, i.e., in 2010.

Figure A.1 shows the employment share in crop cultivation (left panel) and the average agricultural annual income (right panel) by mesoregion in 2010. The employment share in crop cultivation is highest in the North and Northeast regions, while the average agricultural wage is highest in the South and Southeast regions. I calculate annual agricultural income by multiplying the reported total monthly income of agricultural workers by 12.

FIGURE A.1
Employment share in crop cultivation and agricultural income by mesoregion, 2010

(A) Employment share in crop cultivation (B) Annual agricultural income by mesoregion

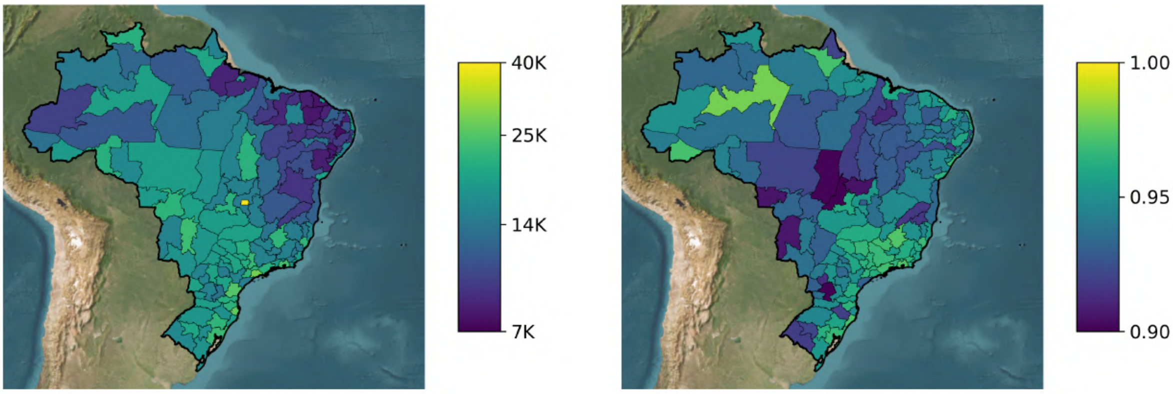


Notes. Panel (a) shows the employment share in crop cultivation by mesoregion in 2010, computed as the number of individuals working in crop cultivation divided by the total number of employed individuals in each mesoregion. Panel (b) shows the average agricultural wage by mesoregion in 2010.

Figure A.2 shows the average annual income (left panel) and the share of respondents who report to have lived in the same mesoregion in 2005 and 2010 (right panel) by mesoregion. The average annual income is highest in the South and Southeast regions, while the emigration rate is highest in the Central regions of Brazil.

FIGURE A.2
Average annual income and emigration rate (2005-2010) by mesoregion

(A) Average annual income by mesoregion (B) Share of non-movers by mesoregion



Notes. Panel (a) shows the average annual income by mesoregion in 2010. Panel (b) shows the share of respondents who report to have lived in the same mesoregion in 2005 and 2010.

A.2. Agricultural production data

The primary data source for agricultural production is the *Produção Agrícola Municipal* (PAM), a survey conducted annually by the Brazilian Institute of Geography and Statistics (IBGE). The PAM provides detailed information on the area planted or destined for harvest, harvested area, physical production quantities, and the nominal value of production for a wide range of temporary and permanent crops at the municipality level.

For the empirical analysis, I use PAM data for the years 1994-2014. I restrict attention to crops with strictly positive reported cultivated area in 2010 and exclude crops that display implausible jumps in yield between 2000 and 2001, ensuring a consistent set of crops across all years of the analysis. The total set of crops is: Soybean (grain), Sugarcane, Maize (grain), Coffee (bean) Total, Paddy rice, Cassava, Beans (grain), Herbaceous cotton (seed cotton), Tobacco (leaf), Tomato, Potato, Grape, Pineapple, Cocoa (bean), Papaya, Onion, Coconut, Rubber (coagulated latex), Garlic, Sorghum (grain), Black pepper, Melon, Sweet potato, Peanut (in shell), Yerba mate (green leaf), Oil palm (fruit bunch), Cashew nut, Sisal or agave (fiber), Heart of palm, Oat (grain), Barley (grain), Castor bean, Annatto (seed), Fava bean, Guarana (seed), Walnut (nut), Mallow (fiber), Tea (green leaf), Pea (grain), Flaxseed, Rye (grain), Ramie (fiber), Jute (fiber), Tung nut.

Table A.4 shows the major crops in Brazil by area planted, quantity produced, and value of production for the period 1994-2014. By area planted, the most important crops are Soybean (33.8%), Maize (24.2%), and Sugarcane (12.0%). By value of production, the most important crops are Soybean (28.4%), Sugarcane (18.5%), and Maize (12.9%).

While Soybeans and Maize are primarily used for either human consumption or animal feed, Sugarcane is predominantly used for biofuel production (ethanol) and sugar.

TABLE A.4
Major crops in Brazil, 1994-2014

Crop	Area Planted (%)	Quantity (%)	Value of Production (%)
Soybean (grain)	33.8	7.6	28.4
Sugarcane	12.0	75.4	18.5
Maize (grain)	24.2	7.2	12.9
Coffee (bean) Total	3.9	0.4	8.0
Paddy rice	5.8	1.7	4.9
Cassava	3.2	3.6	4.9
Beans (grain)	7.5	0.5	3.7
Herbaceous cotton (seed cotton)	1.7	0.4	3.6
Tobacco (leaf)	0.7	0.1	3.0
Tomato	0.1	0.5	2.1
Potato	0.3	0.5	1.9
Grape	0.1	0.2	1.3
Cocoa (bean)	1.2	0.0	0.8
Sorghum (grain)	1.1	0.2	0.3
Cashew nut	1.2	0.0	0.1

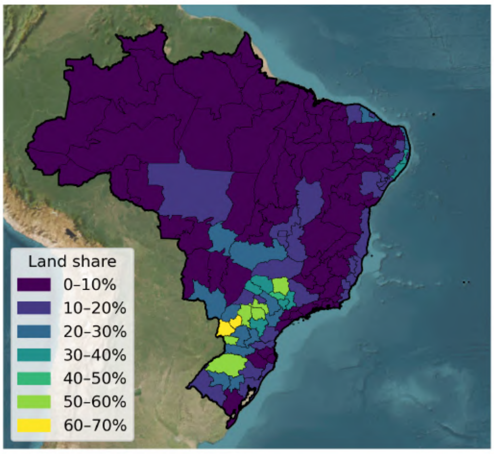
Notes. The table shows the major crops in Brazil by area planted, quantity produced, and value of production for the period 1994-2014. The percentages are calculated as the share of each crop in the total area planted, total quantity produced, and total value of production over the whole period, respectively.

Since the empirical analysis is conducted at the mesoregion level, I aggregate the municipality-level PAM data using official IBGE correspondence tables. I merge the resulting dataset with land-area information (from mesoregion shapefiles by the IBGE) to compute the share of total land devoted to agriculture in each mesoregion-year.

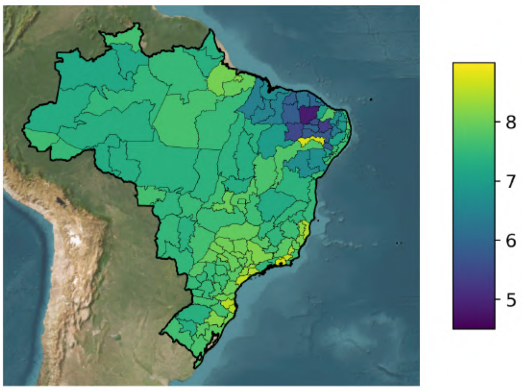
To obtain a measure of constant-price agricultural output per hectare, I revalue crop-level production using reference prices. Specifically, I construct crop-specific reference prices as the ratio of the total value to the total quantity produced over the full sample. Each crop's production is then revalued at these constant prices and aggregated across crops to obtain total output at the mesoregion level. Dividing by the area planted yields an index of agricultural land productivity at constant prices, which forms the basis of the productivity analysis presented in the main text. Figure A.3 shows the share of land used in agriculture, measured agricultural productivity, and fundamental agricultural productivity by mesoregion in 2010.

FIGURE A.3
Land use, measured productivity, and fundamental productivity by mesoregion, 2010

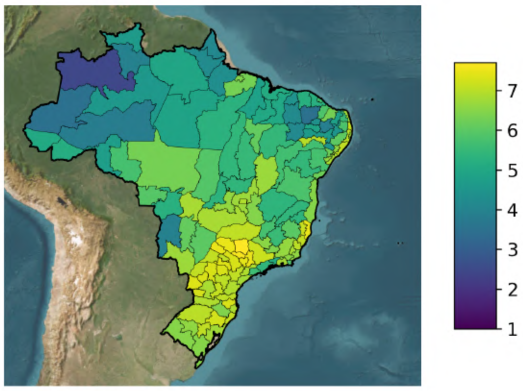
(A) Share of land used in agriculture



(B) Measured agricultural productivity



(C) Fundamental agricultural productivity



Notes. Panel (a) shows the share of land used in agriculture by mesoregion in 2010, computed as the area planted for all crops divided by the total land area in each mesoregion. Panel (b) shows measured agricultural productivity, defined as constant-price output per hectare of land. Panel (c) shows fundamental agricultural productivity, defined as constant-price output per hectare of land adjusted for land use intensity with $\theta = 2$.

Fundamental agricultural productivity is defined as constant-price output per hectare of land adjusted for land use intensity with $\theta = 2$, following the model structure. In the "raw" measured agricultural productivity, where land use intensity is not accounted for, some of the highest agricultural productivity levels are observed in urban areas, including Metropolitana do Rio de Janeiro. This is driven by the fact, that these use a small share of their land for agriculture, but the land that is used is highly productive. Once land use intensity is accounted for, fundamental agricultural productivity is highest in the South and Southeast regions, which aligns much more closely with observed agricultural wages in these regions. This validates that the model-based adjustment for land use intensity is not merely a theoretical construct, but has empirical relevance.

B. Empirical evidence

B.1. Construction of Predicted Counterfactual Incomes

The motivating evidence in Section 3 is based on the 2010 Brazilian Census, which reports a respondents' current residence (in 2010) and residence in 2005. The sample consists of Brazilian-born individuals aged 25-64 with positive income and non-missing information on occupation, place of residence in 2005, race, education, marital status, and gender, yielding roughly 6.2 million observations. Using the data, I estimate a Mincer regression, for each mesoregion j, of the form

$$\log \text{Income}_{ijt} = \mathbf{X}_{ijt}\boldsymbol{\beta}_j + \varepsilon_{ijt}, \tag{19}$$

where i indexes individuals, $t = 2010$, and $\mathbf{X}_{ijt}$ includes age, age squared, gender, race, marital status, and education.

Using the estimated coefficients $\hat{\beta}_j$ from each mesoregion j, I predict counterfactual incomes for all individuals observed in 2010 as if they had remained in their 2005 mesoregion of residence. For the sample of individuals who did not move, I compare predicted and actual incomes to assess the fit of the Mincer regression. The results are shown in Table B.5, which indicates that the Mincer regression explains a substantial portion of income variation, with an R^2 of 0.43.

TABLE B.5
Predicted vs. actual income for non-movers

	Log Actual Income (1)
Constant	6.45×10^{-10}
	(1×10^{-6})
Predicted Log Income	1.000^{***}
	(1×10^{-6})
Observations	5,976,627
R^2	0.43307

Notes. The table shows the results from regressing actual log income on predicted log income for individuals who did not move between 2005 and 2010. The dependent variable is the log of actual income in 2010, and the independent variable is the log of predicted income in 2010 based on the Mincer regression estimated in each mesoregion.

Using the estimated $\hat{\beta}_j$, I then predict 2010 log incomes for individuals observed in 2005 as if they had stayed in their origin mesoregion. I interpret the fitted values from the Mincer regression as counterfactual incomes that individuals would have earned, in expectation, had they remained in their origin mesoregion. Predicted values are exponentiated and grouped into ten 5,000 R$ bins up to 50,000 R$.

B.2. Climate change and agricultural productivity

To establish the link between agricultural productivity and wages, I regress log agricultural wages on log fundamental agricultural productivity in Brazilian mesoregions in 2010. The underlying data is shown in Figure II (panel b) and Figure A.3 (panel c). The regression results are shown in Table B.6, which indicates a strong positive relationship between agricultural productivity and wages, with an elasticity of approximately 0.215 (without fixed effects) and 0.128 with state fixed effects.

TABLE B.6
Relationship between agricultural wages and fundamental productivity (2010)

	Log of mean agricultural income	
	(1)	(2)
Log fundamental productivity	0.2147***	0.1276***
	(0.0226)	(0.0192)
Constant	7.314***	
	(0.1388)	
State fixed effects		✓
Observations	137	136
R^2	0.40001	0.88288
Within R^2		0.28730

Notes. The table shows the results from regressing log agricultural wages on log fundamental agricultural productivity in Brazilian mesoregions in 2010. The dependent variable is the log of the average agricultural wage in each mesoregion. The independent variable is the log of fundamental agricultural productivity, defined as constant-price output per hectare of land adjusted for land use intensity with $\theta = 2$.

C. Derivations

This part describes how the static problem is set up and how the endogenous objects depend on the allocation of labor in space.

C.1. Consumption

Households get utility from consuming a composite agricultural good and a non-agricultural good:

$$C_t^i = \frac{\left(C_t^{i;a}\right)^{\gamma}\left(C_t^{i;n}\right)^{1-\gamma}}{\gamma^{\gamma}(1-\gamma)^{1-\gamma}},$$

where $C_t^{i;a}$ denotes consumption of the composite agricultural good and $C_t^{i;n}$ denotes consumption of the non-agricultural good. γ is the constant expenditure share on the agricultural good (Cobb-Douglas).

The composite agricultural good is a constant elasticity of substitution (CES) aggregate of regional varieties:

$$C_t^{i,a} = \left(\sum_{j \in \mathcal{S}} \left(\psi^j \right)^{1/\epsilon} \left(c_t^{j,i;a} \right)^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}},$$

where $c_t^{j,i;a}$ is the consumption of the regional variety of the agricultural good from origin j. ϵ is the elasticity of substitution between different varieties of the agricultural good (Armington elasticity). ψ^j is a preference (or quality) parameter for the agricultural good from origin j that is constant over time and subject to the normalization $\sum_{j \in \mathcal{S}} \psi^j = 1$.[1] As in Costinot et al. (2016), the non-agricultural good is freely traded, and it is used as the numeraire for the remainder of the paper.

The final consumption bundle satisfies the budget constraint:

$$P_t^{i;a} C_t^{i;a} + p_t^0 C_t^{i;n} \leq y_t^i,$$

where $P_t^{i;a}$ is the CES price index and p_t^0 is the price of the non-agricultural good at time t (normalized to one in the initial period).[2,3] The regional price index for final consumption is:

$$P_t^i = \left(P_t^{i;a} \right)^{\gamma} \left(p_t^0 \right)^{1-\gamma}.$$

C.2. Production technology

The economy consists of two sectors: agriculture and non-agriculture. The agricultural sector utilizes land and labor, whereas the non-agricultural sector employs labor only. Each region i is endowed with X^i units of land and L_t^i units of labor. Labor is perfectly mobile across sectors within regions, but not across regions.

For the agricultural production technology, I closely follow Costinot et al. (2016); Gollin and Wolfersberger (2024); Sotelo (2020) and assume that land consists of a continuum of heterogeneous plots ω in each region i that differ in their productivity and fixed costs to operate.[4]

[1] Because I lack intra-national trade flow data, I assume ψ^j to be origin-specific, rather than origin-destination specific (as in Costinot et al., 2016). I set this parameter to clear the agricultural goods market in the calibration. In a planned extension, where $\mathcal{S}$ includes foreign countries, I allow this parameter to vary between country pairs to match country-to-country trade flows.

[2] The CES price index is given by

$$P_t^{i;a} = \left(\sum_{j \in \mathcal{S}} \left(\psi^j \right)^{1/\epsilon} \left(p_t^{j,i;a} \right)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}},$$

where $p_t^{j,i;a}$ is the price of the agricultural variety produced in j in destination market i.

[3] In the counterfactual scenarios, p_t^0 adjusts to guarantee market clearing and must not be one.

[4] This fixed cost is needed to reconcile the fact that not all land is used for the cultivation of crops. If land were costless, agricultural producers would utilize all land for the cultivation of crops. Incomplete specialization of labor is guaranteed by the outside option for labor: working in non-agriculture.

Agricultural firms make profits that are spent locally on consumption. The costs incurred for operating a plot (e.g., slashing vegetation or clearing forest) are likewise spent locally on consumption.

The non-agricultural sector consists of a continuum of jobs η that differ in productivity and wages. Each job in region i produces a homogeneous good using labor, with productivity varying across jobs and regions.[5] The sector is perfectly competitive.

Agriculture Production technology in agriculture is Cobb-Douglas, using land and labor:

$$q_t^i(\omega) = \left[L_t^i(\omega)\right]^{\alpha} \left[z_t^i(\omega) X_t^i(\omega)\right]^{1-\alpha},$$

where $q_t^i(\omega)$ is the output of plot ω in region i at time t, $L_t^i(\omega)$ is the amount of labor used on plot ω in region i at time t, and $X_t^i(\omega)$ is the amount of land used on plot ω in region i at time t. The parameter α represents the share of labor in production. $z_t^i(\omega)$ is the plot's productivity.

As in Gollin and Wolfersberger (2024), a representative farmer must pay a monetary cost $z^{i0}(\omega)$ expressed in units of the non-agricultural good to cultivate a plot of land. This cost needs to be paid every period a plot is used.

Productivity $z_t^i(\omega)$ and opening costs $z^{i0}(\omega)$ are randomly distributed across plots ω in region i and drawn from a nested Fréchet distribution with parameters $\{A_t^i, F^{i0}, \theta\}$. Formally, the distribution of land quality $z_t^i(\omega)$ and investment in land $z^{i0}(\omega)$ is given by:

$$\Pr\left(z_t^i(\omega) \leq z_t^i, z^{i0}(\omega) \leq z^{i0}\right) = \exp\left(-o\left(\left(z_t^i / A_t^i\right)^{-\theta} + \left(z^{i0} / F^{i0}\right)^{-\theta}\right)\right),$$

where $\theta > 1$ is the shape parameter of the distribution. o is a normalization constant that is set such that $A_t^i = E[z_t^i(\omega)]$ and $F^{i0} = E[z^{i0}(\omega)]$. A_t^i and F^{i0} are the *unconditional* average land quality and initial investment in land across all plots in region i at time t.[6]

Non-agriculture The non-agricultural production function for a job η in region i is given by:

$$q_t^{i;n}(\eta) = A^{i;n}(\eta) L_t^{i;n}(\eta),$$

where $q_t^{i;n}(\eta)$ is the output of job η in region i at time t, $A^{i;n}(\eta)$ is the productivity of job η in region i, and $L_t^{i;n}(\eta)$ is the amount of labor used in job η in region i at time t.

Productivity $A^{i;n}(\eta)$ is drawn from a log-normal distribution with parameters μ^i and σ^i, which captures the heterogeneity in job productivity across potential jobs in non-agriculture in region i:

$$\ln A^{i;n}(\eta) \sim \mathcal{N}(\mu^i, (\sigma^i)^2).$$

[5] This structure allows me to capture the stark income inequality within the non-agricultural sector across Brazilian regions, unlike a representative-agent framework where a single wage would prevail.

[6] Formally, ξ is given by: $\xi = \Gamma\left(1 + \frac{1}{\theta}\right)^{-\theta}$, where $\Gamma(\cdot)$ is the gamma function.

C.3. Temporary equilibrium

Following Caliendo et al. (2019), I refer to the equilibrium of the static model as a *temporary equilibrium*. In this temporary equilibrium, the allocation of labor across regions is fixed, and the model determines the equilibrium prices and wages in each region such that all markets clear.

Labor income & allocation Households are randomly matched to jobs in the non-agricultural sector. Each household receives a job offer associated with income $A^{i;n}(\eta)$, where η is a random draw from the distribution of job offers in region i. The household then compares this offer to the agricultural wage $w_t^{i;a}$, which serves as a reservation wage. If the job offer is below the agricultural wage, the household chooses to work in agriculture; otherwise, they accept the job offer in the non-agricultural sector.

The fraction working in agriculture is

$$\lambda_t^i = \Pr\big(A^{i,n}(\eta) < w_t^{i;a}\big) = \Phi\Big(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\Big),$$

where $\Phi(\cdot)$ is the standard normal cumulative density function (CDF). Agricultural employment is $L_t^{i;a} = \lambda_t^i L_t^i$.

The resulting income distribution, $f_t^i(y)$, is a truncated log-normal distribution with a point mass at the agricultural wage $w_t^{i;a}$:

$$f_t^i(y) = \lambda_t^i \delta\big(y - w_t^{i;a}\big) + (1-\lambda_t^i) \cdot \frac{1}{y\sigma^i\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu^i)^2}{2(\sigma^i)^2}\right) \cdot \frac{\mathbb{I}\big(y \geq w_t^{i;a}\big)}{1 - \Phi\left(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\right)}, \tag{20}$$

where $\delta(\cdot)$ is the Dirac delta function and $\mathbb{I}(\cdot)$ is the indicator function. The income distribution is time-varying and depends on the agricultural wage $w_t^{i;a}$.

The average non-agricultural income among those who accept a non-agricultural productivity draw is[7]

$$w_t^{i;n} \equiv E\big[y \mid y \geq w_t^{i;a}\big] = \exp\Big(\mu^i + \tfrac{1}{2}(\sigma^i)^2\Big)\ \frac{1 - \Phi\big(\frac{\ln w_t^{i;a} - \mu^i - (\sigma^i)^2}{\sigma^i}\big)}{1 - \Phi\big(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\big)}.$$

Agriculture In each region, agricultural firms choose the amount of land to use in agriculture $X_t^i(\omega)$ and the amount of labor to use in agriculture $L_t^i(\omega)$ to maximize profits.

The firms must pay a monetary cost $z^{i0}(\omega)$ to use land in agriculture, which is expressed in units of the non-agricultural good. Agricultural firms are maximizing:

$$\max_{L_t^i(\omega), X_t^i(\omega)} \quad \pi_t^i(\omega) = p_t^{i;a} q_t^i(\omega) - w_t^{i;a} L_t^i(\omega) - p_t^0 z^{i0}(\omega) X_t^i(\omega).$$

[7] This setup has an appealing implication. When regional agricultural wages decline, non-agricultural wages tend to fall on average, but their variance increases. Hence, climate change, or a decline in agricultural incomes in general, leads to an increase in income inequality within a region. Both are driven by agricultural versus non-agricultural incomes, as well as by increasing inequality within non-agricultural incomes.

The first-order condition with respect to labor yields:

$$w_t^{i;a} = \alpha p_t^{i;a} \left[\frac{z_t^i(\omega) X_t^i(\omega)}{L_t^i(\omega)} \right]^{1-\alpha}.$$

Rearranging for labor and substituting back into the profit function allows to write:

$$\pi_t^i(\omega) = X_t^i(\omega) \left[z_t^i(\omega)\Omega_t^i - p_t^0 z^{i0}(\omega) \right], \tag{21}$$

with $\Omega_t^i \equiv \left(\alpha^\alpha (1-\alpha)^{1-\alpha} p_t^{i;a} \left(w_t^{i;a} \right)^{-\alpha} \right)^{\frac{1}{1-\alpha}}$.

This expression indicates that profit is linear with respect to the amount of land used. Hence, the optimal land use decision for each plot is a corner solution: either the firm sets $X_t^i(\omega) = 1$ (and operates the plot) or $X_t^i(\omega) = 0$ (and leaves the plot unused).

From eq. 21 (as in Gollin and Wolfersberger, 2024), the implicit net return to land is

$$r_t^i(\omega) = z_t^i(\omega)\Omega_t^i - p_t^0 z^{i0}(\omega).$$

This implicit return reflects the net revenue generated by the land after paying labor and monetary costs. It is: i) strictly positive for inframarginal plots (which are operated and yield profits), ii) zero for the marginal plot (which breaks even), iii) negative for unused plots (which are not worth operating at current prices).

Together with the distributional assumption on productivity, the fraction of land that is profitable to be cultivated, and hence, used in agriculture in region i at time t is

$$\begin{aligned} \zeta_t^i &\equiv \Pr\left(z_t^i(\omega)\Omega_t^i \geq \max\left\{ p_t^0 z^{i0}(\omega), z_t^i(\omega)\Omega_t^i \right\} \right) \\ &= \frac{\left(A_t^i \Omega_t^i\right)^\theta}{\left(A_t^i \Omega_t^i\right)^\theta + \left(F^{i0} p_t^0\right)^\theta}. \end{aligned} \tag{22}$$

This equation shows that the share of land used in agriculture in region i increases in agricultural profitability, $A_t^i \Omega_t^i$, and decreases in the monetary cost of making land available for agriculture, $F^{i0} p_t^0$.

Average productivity is endogenous and depends on the fraction of land used. In particular, farmers will first exploit plots with high productivity and low operational costs.[8]

$$E[z_t^i(\omega)|\omega \text{ optimal}] = A_t^i \left(\zeta_t^i\right)^{-1/\theta}, \quad E[z^{i0}(\omega)|\omega \text{ optimal}] = F^{i0} \left(1 - \zeta_t^i\right)^{1/\theta}, \tag{23}$$

where $E[z_t^i(\omega)|\omega \text{ optimal}]$ is the expected productivity of the plots that are cultivated, and $E[z^{i0}(\omega)|\omega \text{ optimal}]$ is the expected cost of cultivating a plot.

[8] In general, the average productivity of agricultural land is larger than the unconditional average productivity of all plots in a region unless all land is cultivated ($\zeta_t^i = 1$). This also implies that the measured productivity described in **??** will not reflect A_t^i, but rather $A_t^i \left(\zeta_t^i\right)^{-1/\theta}$.

The output is thus:

$$Q_t^{i;a} = \left(L_t^{i;a}\right)^{\alpha} \left(A_t^i \left(\zeta_t^i\right)^{-1/\theta} X_t^{i;a}\right)^{1-\alpha}, \tag{24}$$

where $L_t^{i;a} = \lambda_t^i L_t^i$ and $X_t^{i;a} = \zeta_t^i X_t^i$.

The sum of fixed costs, B_t^i, and total profits of agricultural firms, R_t^i, are:

$$B_t^i = E[z^{i0}(\omega)|\omega \text{ optimal}] \cdot \zeta_t^i X_t^i = F^{i0}(1-\zeta_t^i)^{1/\theta}\zeta_t^i X_t^i \tag{25}$$

$$R_t^i = p_t^i Q_t^{i;a} - \left(w_t^{i;a} L_t^{i;a}\right) - \left(F^{i0}(1-\zeta_t^i)^{1/\theta}\zeta_t^i X_t^i\right). \tag{26}$$

Wages are equal to the marginal product of labor on the marginal plot:

$$p_t^{i;a}\frac{\partial Q_t^{i;a}}{\partial L_t^{i;a}} = w_t^{i;a}$$

$$w_t^{i;a} = p_t^{i;a}\alpha\left(\frac{A_t^i\left(\zeta_t^i\right)^{-1/\theta} X_t^{i;a}}{L_t^{i;a}}\right)^{1-\alpha}. \tag{27}$$

Non-agriculture The non-agricultural production is implicitly determined by the agricultural wage. Only jobs that have a productivity $A^{i;n}(\eta)$ above the agricultural wage $w_t^{i;a}$ are filled.

Total output in the non-agricultural sector is then given by:

$$Q_t^{i;n} = \int_{A^{i;n}(\eta)\geq w_t^{i;a}} A^{i;n}(\eta) L_t^{i;n}(\eta) d\eta, \tag{28}$$

where $L_t^{i;n}(\eta)$ is the amount of labor used in job η in region i at time t. Without loss of generality, the amount of labor used in a job η can be normalized to one. The total amount of labor in the non-agricultural sector is given by: $L_t^{i;n} = (1-\lambda_t^i)L_t^i$. Hence, eq. 28 reduces to:

$$Q_t^{i;n} = w_t^{i;n}(1-\lambda_t^i)L_t^i, \tag{29}$$

which shows that the total output in the non-agricultural sector is equal to the average non-agricultural income $w_t^{i;n}$ times the amount of labor in the non-agricultural sector $(1-\lambda_t^i)L_t^i$.

Market clearing The sum of the wage bill, profits, and land-clearing costs gives total income in a region i at time t:

$$Y_t^i = \underbrace{\left(y_t^i L_t^i\right)}_{\text{Wage bill}} + \underbrace{\left(R_t^i\right)}_{\text{Profits}} + \underbrace{\left(B_t^i\right)}_{\text{Land-clearing}}, \tag{30}$$

where y_t^i is the average income in region i at time t, which is given by:

$$y_t^i = \lambda_t^i w_t^{i;a} + (1-\lambda_t^i) w_t^{i;n}.$$

What remains to be done is to derive the trade between regions. The Cobb-Douglas preferences over the composite agricultural and the non-agricultural good imply expenditure shares of γ and $1-\gamma$, respectively. Within agricultural expenditures, the expenditure share on variety j is:

$$\pi_t^{j,i;a} = \frac{\psi^j \left(p_t^{j;a}\delta^{j,i}\right)^{1-\epsilon}}{\sum_{k\in\mathcal{S}} \psi^k \left(p_t^{k;a}\delta^{k,i}\right)^{1-\epsilon}}, \tag{31}$$

where $\delta^{j,i}$ are the iceberg trade costs between j and i.

Let $S_t^{j;a} \equiv p_t^{j;a} Q_t^{j;a}$ be the value of agricultural production of region j at time t.[9] Then, market clearing requires:

$$S_t^{j;a} = \gamma \sum_{i\in\mathcal{S}} \pi_t^{j,i;a} Y_t^i. \tag{32}$$

Non-agricultural market clearing is achieved when the economy-wide value of non-agricultural production is equal to the demand for non-agriculture:

$$\sum_{j\in\mathcal{S}} S_t^{j;n} = (1-\gamma) \sum_{i\in\mathcal{S}} Y_t^i. \tag{33}$$

D. Exact-hat derivations

This section presents the hat algebra system used to solve for counterfactual equilibria in the spatial model. By expressing equilibrium conditions in proportional changes relative to the baseline, the system captures how wages, land use, sectoral labor allocation, trade shares, and market clearing change in response to changes in the agricultural productivity $\hat{A}_t^i$ and changes in the allocation of labor across regions $\hat{L}_t^i$. Since the dynamic model requires real wage levels, these are recovered ex post from baseline values and proportional changes.

- Agricultural wages:

$$\begin{aligned} w_t^{i;a} &= p_t^{i;a}\alpha \left(\frac{A_t^i \left(\zeta_t^i\right)^{-1/\theta} X_t^{i;a}}{L_t^{i;a}}\right)^{1-\alpha} \\ &= \alpha \left(X_t^i\right)^{1-\alpha} p_t^{i;a} \left(A_t^i\right)^{1-\alpha} \left(\zeta_t^i\right)^{\frac{\theta-1}{\theta}(1-\alpha)} \left(\lambda_t^i\right)^{\alpha-1} \left(L_t^i\right)^{\alpha-1} \end{aligned}$$

  where α and X_t^i are constant.

  In changes:

$$\hat{w}_t^{i;a} = \left(\hat{p}_t^{i;a}\right) \left(\hat{A}_t^i\right)^{1-\alpha} \left(\hat{\zeta}_t^i\right)^{\frac{(\theta-1)(1-\alpha)}{\theta}} \left(\hat{\lambda}_t^i\right)^{\alpha-1} \left(\hat{L}_t^i\right)^{\alpha-1}$$

[9] I use S for "sales" to avoid confusion with land X.

- Land use:

$$\zeta_t^i = \frac{\left(A_t^i \Omega_t^i\right)^\theta}{\left(A_t^i \Omega_t^i\right)^\theta + \left(F_t^{i0} p_t^{i0}\right)^\theta}$$

where F_t^{i0} is a constant.

In changes:

$$\hat{\zeta}_t^i = \frac{\left(\hat{A}_t^i \hat{\Omega}_t^i\right)^\theta}{\zeta_t^i \left(\hat{A}_t^i \hat{\Omega}_t^i\right)^\theta + \left(1 - \zeta_t^i\right) \left(\hat{p}_t^{i0}\right)^\theta}$$

- Net profitability:

$$\Omega_t^i = \left(\tilde{\alpha} \frac{p_t^{i;a}}{\left(w_t^{i;a}\right)^\alpha}\right)^{\frac{1}{1-\alpha}}$$

In changes:

$$\hat{\Omega}_t^i = \left(\hat{p}_t^{i;a} \left(\hat{w}_t^{i;a}\right)^{-\alpha}\right)^{\frac{1}{1-\alpha}}$$

- Sectoral labor allocation:

$$\lambda_t^i = \Phi\left(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\right)$$

in changes:

$$\hat{\lambda}_t^i = \frac{\Phi\left(\frac{\ln\left(w_t^{i;a} \hat{w}_t^{i;a}\right) - \mu^i}{\sigma^i}\right)}{\Phi\left(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\right)}$$

- Trade shares:

$$\pi_t^{j,i;a} = \frac{\psi^j \left(p_t^{j;a} \kappa^{j,i}\right)^{1-\epsilon}}{\sum_k \psi^k \left(p_t^{k;a} \kappa^{k,i}\right)^{1-\epsilon}}$$

where κ is constant.

In changes:

$$\hat{\pi}_t^{j,i;a} = \frac{\left(\hat{p}_t^{j;a}\right)^{1-\epsilon}}{\sum_k \pi_t^{k,i;a} \left(\hat{p}_t^{k;a}\right)^{1-\epsilon}}$$

- Agricultural market clearing:

$$X_t^{j;a} = \gamma \sum_i \pi_t^{j,i;a} Y_t^i$$

In changes:

$$\hat{X}_t^{j;a} = \frac{\sum_i \hat{\pi}_t^{j,i;a} \hat{Y}_t^i \pi_t^{j,i;a} Y_t^i}{\sum_i \pi_t^{j,i;a} Y_t^i}$$

- Labor market clearing (non-agricultural wage):

$$w_t^{i;n} = \exp\left(\mu^i + \frac{1}{2}(\sigma^i)^2\right) \frac{1 - \Phi\left(\frac{\ln w_t^{i;a} - \mu^i - (\sigma^i)^2}{\sigma^i}\right)}{1 - \Phi\left(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\right)}$$

In changes:

$$\hat{w}_t^{i;n} = \frac{1 - \Phi\left(\frac{\ln w_t^{i;a} + \ln \hat{w}_t^{i;a} - \mu^i - (\sigma^i)^2}{\sigma^i}\right)}{1 - \Phi\left(\frac{\ln w_t^{i;a} + \ln \hat{w}_t^{i;a} - \mu^i}{\sigma^i}\right)} \frac{1 - \Phi\left(\frac{\ln w_t^{i;a} - \mu^i}{\sigma^i}\right)}{1 - \Phi\left(\frac{\ln w_t^{i;a} - \mu^i - (\sigma^i)^2}{\sigma^i}\right)}$$

- Total income:

$$Y_t^i = w_t^{i;a} L_t^{i;a} + w_t^{i;n} L_t^{i;n} + R_t^i + B_t^i$$

In changes:

$$\hat{Y}_t^i = \frac{\hat{w}_t^{i;a} w_t^{i;a} \hat{\lambda}_t^i \lambda_t^i \hat{L}_t^i L_t^i + \hat{w}_t^{i;n} w_t^{i;n} (1 - \hat{\lambda}_t^i \lambda_t^i) \hat{L}_t^i L_t^i + \hat{R}_t^i R_t^i + \hat{B}_t^i B_t^i}{Y_t^i}$$

- Quantity produced in agriculture:

$$Q_t^{i;a} = \left(\lambda_t^i L_t^i\right)^{\alpha} \left(A_t^i \left(\zeta_t^i\right)^{\frac{\theta-1}{\theta}} X_t^i\right)^{1-\alpha}$$

where X_t^i is constant.

In changes:

$$\hat{Q}_t^{i;a} = \left(\hat{\lambda}_t^i \hat{L}_t^i\right)^{\alpha} \left(\hat{A}_t^i \left(\hat{\zeta}_t^i\right)^{\frac{\theta-1}{\theta}}\right)^{1-\alpha}$$

- Agricultural production value (in changes):

$$\hat{p}_t^{i;a} \hat{Q}_t^{i;a}$$

which must, in the counterfactual equilibrium, be equal to $\hat{X}_t^{i;a}$ for markets to clear. I.e., we have the counterfactual agricultural goods market-clearing condition:

$$\hat{p}_t^{j;a} = \frac{\hat{X}_t^{j;a}}{\hat{Q}_t^{j;a}}$$

$$\hat{p}_t^{j;a} = \frac{\frac{\sum_i \hat{\pi}_t^{j,i;a} \hat{Y}_t^i \pi_t^{j,i;a} Y_t^i}{\sum_i \pi_t^{j,i;a} Y_t^i}}{\hat{Q}_t^{j;a}}$$

- Non-agricultural market clearing:

$$\hat{p}_t^{j0} = \frac{\hat{X}_t^{j;n}}{\hat{Q}_t^{j;n}}$$

- Profits:

$$R_t^i = p_t^{i;a} Q_t^{i;a} - w_t^{i;a} L_t^{i;a} - B_t^i$$

In changes:

$$\hat{R}_t^i = \frac{\hat{p}_t^{i;a} \hat{Q}_t^{i;a} p_t^{i;a} Q_t^{i;a} - \hat{w}_t^{i;a} \hat{\lambda}_t^i \hat{L}_t^i w_t^{i;a} \lambda_t^i L_t^i - \hat{B}_t^i B_t^i}{R_t^i}$$

- Income from fixed costs:

$$B_t^i = F_t^{i0} \left(1 - \zeta_t^i\right)^{1/\theta} \zeta_t^i X_t^i$$

where F_t^{i0} and X_t^i are constants.

In changes:

$$\hat{B}_t^i = \left(\frac{1 - \hat{\zeta}_t^i \zeta_t^i}{1 - \zeta_t^i}\right)^{1/\theta} \hat{\zeta}_t^i$$

The algorithm to solve the general equilibrium of the static model nests two main loops. In the *outer loop*, I update the vector of agricultural producer-price changes $\{\hat{p}^{j;a}\}$.

Wrapped inside is an *inner loop* over regions. For each region i, I treat the current $\hat{p}^{i;a}$ as given and solve for the corresponding change in the agricultural wage $\hat{w}^{i;a}$ by finding the root of

$$\hat{w} \; - \; M^i(\hat{w}) \; = \; 0,$$

where

$$M^i(\hat{w}) = \hat{p}^{i;a} \, (\hat{A}^i)^{1-\alpha} \left[\hat{\zeta}^i(\hat{w})\right]^{\frac{(\theta-1)(1-\alpha)}{\theta}} \left[\hat{\lambda}^i(\hat{w})\right]^{\alpha-1} (\hat{L}^i)^{\alpha-1}.$$

Because land-use shares $\hat{\zeta}^i$ and labor-allocation shares $\hat{\lambda}^i$ both depend non-linearly on $\hat{w}^{i;a}$, I solve this one-dimensional fixed-point problem for each i.

Once each $\hat{w}^{i;a}$ is determined, I back out the non-agricultural wage change $\hat{w}^{i;n}$, recompute aggregate income changes $\hat{Y}^i$, and update trade shares $\hat{\pi}^{j,i;a}$. Feeding these into the market-clearing condition gives a new price guess

$$\hat{p}^{j;a}_{\text{new}} = \frac{\sum_i \hat{\pi}^{j,i;a}\, \hat{Y}^i\, \pi^{j,i;a}\, Y^i}{\sum_i \pi^{j,i;a}\, Y^i} \Big/ \hat{Q}^{j;a},$$

which I under-relax as

$$\hat{p}^{j;a} \leftarrow \rho\, \hat{p}^{j;a}_{\text{new}} + (1-\rho)\, \hat{p}^{j;a},$$

to ensure stable convergence.

I repeat both loops until the maximum changes in $\hat{p}^{j;a}$ and $\hat{w}^{i;a}$ fall below the chosen tolerance, yielding the full counterfactual equilibrium in prices, wages, quantities, and trade flows.

E. Calibration details

E.1. Baseline economic environment and calibration results

This section presents additional results from the static calibration. By construction, the model exactly matches key moments in the baseline year 2010, as shown in Table E.7. In particular, by model inversion, the model exactly matches in each region: i) total labor, ii) average wages, iii) the employment share in agriculture, iv) the average agricultural wage, v) the non-agricultural average wages, vi) total land endowments, and vii) the fraction of land used in agriculture.

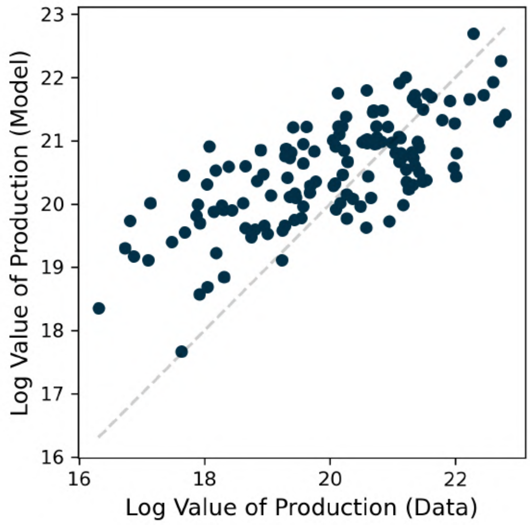
TABLE E.7
Exactly matched moments in 2010

Parameter	Mean	Min	Max
Exactly matched			
Labor:			
L^j	440222	11612	7160612
w^j	15576.17	7444.09	39283.89
λ^j	0.1243	0.0030	0.3193
$w^{j;a}$	5861.50	2813.27	10649.65
$w^{j;n}$	16770.86	8800.74	39554.08
Land:			
X^j	6558425	283872	50469934
ζ^j	0.1321	0.0001	0.6959

Notes: The table shows key moments that the static model exactly matches in the baseline year 2010. I report the unweighted mean across regions, as well as the minimum and maximum values.

In the model calibration, the Cobb-Douglas parameter α is set to match the agricultural wage bill-to-output-ratio. I.e., $\alpha = \frac{\sum_i w^{i;a} L^{i;a}}{\sum_i V_t^{i;a}}$, where $V_t^{i;a}$ is the value of agricultural production in region i reported by the PAM dataset. I calibrate a value of $\alpha = 0.1579$ to match the data. However, because of this it is not strictly necessary that the model also matches the agricultural wage bill-to-output ratio in each region. To match the agricultural wage bill-to-output ratio in each region, I would need to introduce additional heterogeneity in the model, for example by allowing for region-specific α_i. Figure E.4 shows that, under the assumption of one α at the

FIGURE E.4
Model fit: Value of production



Notes: The figure shows data vs. model fit for the log value of agricultural production in 2010. The R^2 of the fit is 0.5.

national level, the model fits the regional agricultural value of production reasonably well, with an R^2 of 0.5.

E.2. Counterfactual productivity paths

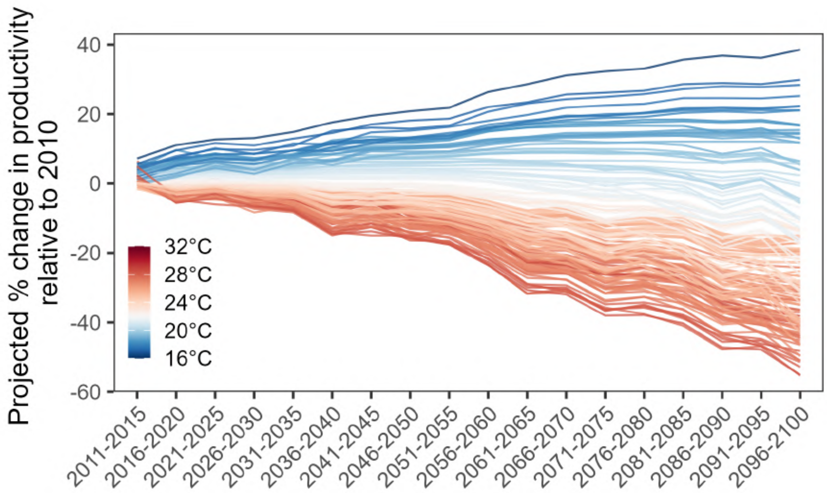
Figure E.5 shows the counterfactual agricultural productivity paths for each region. The paths are constructed by combining the estimated climate impacts on agricultural productivity from Section 4 with the RCP 8.5 temperature projections from the Copernicus Climate Change Service (C3S) CMIP6 archive. I use monthly near-surface air temperature projections from the CMCC-ESM2 Earth System Model.

E.3. Migration costs

Figure E.7 reports model-implied migration probabilities by income, averaged across regions using national population weights. The model successfully reproduces the qualitative pattern that mobility rises with income, but it overstates the gradient relative to the empirical evidence. Low-income households migrate slightly less than observed, while high-income households migrate too frequently. This discrepancy likely reflects a mechanical feature of the extreme-value preference structure adopted from the spatial-equilibrium literature. In the logit formulation, workers draw idiosyncratic location preferences whose variance is governed by the dispersion parameter ν. At high incomes—where expected utility differences across locations are small and pecuniary migration costs are negligible—these random preference draws dominate, generating excess migration in the upper tail. The structure preserves tractability and closed-form choice probabilities, but it likely inflates high-income mobility relative to reality.

FIGURE E.5

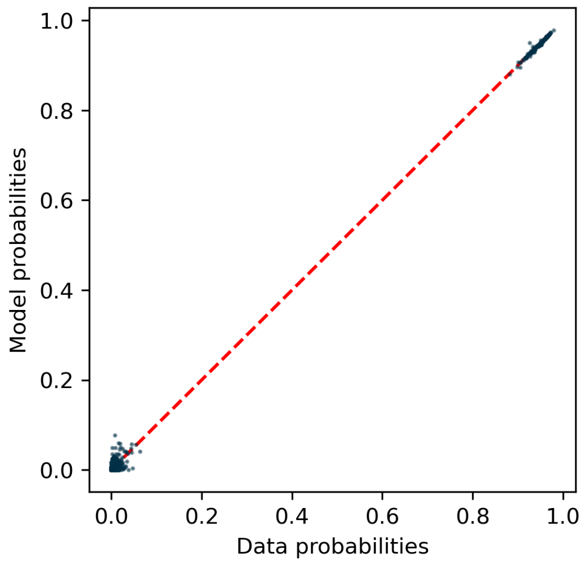
Counterfactual agricultural productivity paths



Notes: The figure shows the counterfactual agricultural productivity paths for each region. The paths are constructed by combining the estimated climate impacts on agricultural productivity from Section 4 with the RCP 8.5 temperature projections from the Copernicus Climate Change Service (C3S) CMIP6 archive. I use monthly near-surface air temperature projections from the CMCC-ESM2 Earth System Model.

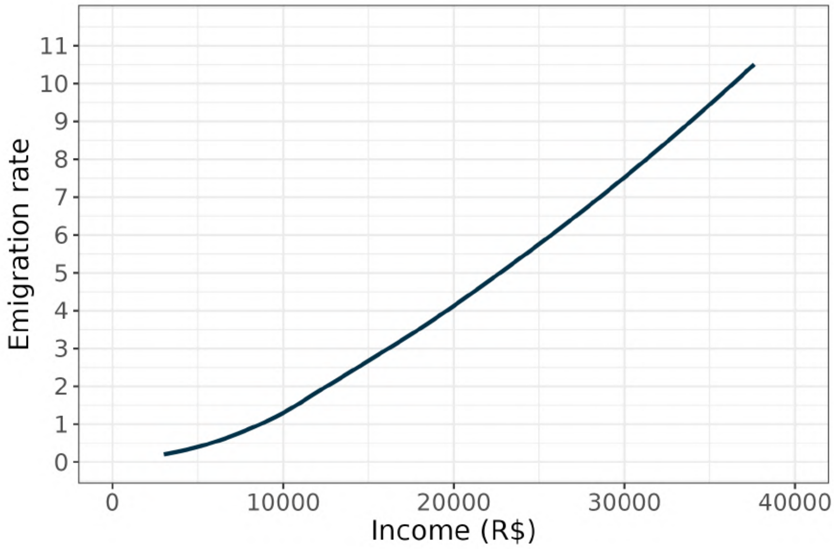
FIGURE E.6

Migration probabilities: Model vs. data



Notes. This figure compares the model-implied and observed bilateral migration probabilities between Brazilian mesoregions. Migration probabilities are defined as $L^{j,i} / \sum_j L^j$, that is, the share of workers from origin j residing in destination i. Observed probabilities are derived from the 2010 Population Census (*Censo Demográfico*) and reflect residential locations in 2005 and 2010. Model-implied probabilities correspond to the first-period migration probabilities generated by the calibrated dynamic spatial model, which infers monetary and non-monetary migration costs by minimizing the distance between simulated and observed probabilities. Monetary costs increase linearly with geographic distance (R$50 per km), and origin-specific non-monetary "home" preferences are calibrated to match regional emigration rates. The resulting specification closely reproduces observed migration patterns, with an overall R^2 above 0.99; focusing on outflows only, the R^2 is approximately 0.45.

FIGURE E.7
Migration probabilities by income (model)



Notes: Model-implied average emigration rate by income, averaged across regions for 2010.

F. Solving the model

To compute the evolution of regional labor allocations, wages, and welfare under forward-looking migration, I solve for the dynamic spatial equilibrium in seven steps. The procedure links the dynamic discrete-choice problem to the static trade equilibrium, ensuring full intertemporal consistency between migration decisions and spatial price systems.

1. **Static equilibrium.** For each period $t \leq T$, and given a conjecture for the regional labor distribution L_t^j, I solve for the *temporary static equilibrium*. This delivers region-specific agricultural wages $w_t^{j,a}$, non-agricultural income distributions $f_t^j(y)$, price indices P_t^j, and sectoral employment shares λ_t^j. The static block is solved using a fixed-point algorithm over agricultural wages and land use, following the derivations in Appendix C.

2. **Income distributions.** Within each region j, income follows a truncated lognormal distribution with parameters (μ^j, σ^j), truncated below by the agricultural wage $w_t^{j,a}$. A mass point λ_t^j at $w_t^{j,a}$ represents agricultural workers. These distributions generate rank-specific expected incomes $y_t^j(r)$ for ranks $r \in (0,1)$, which form the state variable for the dynamic problem.

3. **Rank-specific consumption utility.** For a household of rank r residing in region j, real consumption utility in period t is

$$U_t^j(r) = \log\left(\frac{y_t^j(r)}{P_t^j}\right).$$

   This term enters additively in the lifetime value function.

4. **Dynamic value recursion.** Starting from the terminal period T, households choose migration paths to maximize expected discounted utility:

$$V_t^j(r) = U_t^j(r) + \nu \log \sum_{i \in \mathcal{S}} \exp\left(\frac{\beta\left[\rho V_{t+1}^i(r) + (1-\rho)\bar{V}_{t+1}^i\right] - \tau^{j,i}(y_t^j(r))}{\nu} \right),$$

where $\tau^{j,i}(y_t^j(r)) = \tilde{\tau}^{j,i}(y_t^j(r)) + \mathbf{1}(i \neq j)\kappa^j$ denotes total migration costs. The monetary component is

$$\tilde{\tau}^{j,i}(y) = \log\left(\frac{y}{y - \chi^{j,i}}\right),$$

which is finite only for $y > \chi^{j,i}$. The parameter ρ governs income persistence: with probability ρ the same rank is retained next period, and with probability $1-\rho$ a new rank is drawn uniformly. The terminal condition is purely static,

$$V_T^j(r) = U_T^j(r),$$

eliminating artificial migration incentives from infinite-horizon continuation values.

5. **Migration probabilities.** Given continuation values $V_{t+1}^i(r)$, the probability that a household of rank r in origin j migrates to destination i at the end of period t is

$$\mu_t^{j,i}(r) = \frac{\exp\Big(\big[\beta(\rho V_{t+1}^i(r) + (1-\rho)\bar{V}_{t+1}^i) - \tau^{j,i}(y_t^j(r))\big]/\nu\Big)}{\sum_{k \in \mathcal{S}} \exp\Big(\big[\beta(\rho V_{t+1}^k(r) + (1-\rho)\bar{V}_{t+1}^k) - \tau^{j,k}(y_t^j(r))\big]/\nu\Big)}.$$

Aggregating over income ranks yields total migration flows:

$$\mu_t^{j,i} = \int_0^1 \mu_t^{j,i}(r)\, dr.$$

6. **Labor propagation.** Population dynamics follow from the aggregation of migration flows:

$$L_{t+1}^i = \sum_{j \in \mathcal{S}} \mu_t^{j,i} L_t^j.$$

To account for income persistence, I implement a Calvo-style update: a fraction ρ of households retain their income ranks, while the remainder are re-ranked uniformly within the new destination's income distribution. This preserves equal-mass income bins in each region and ensures that local income distributions remain consistent with the static equilibrium.

7. **Convergence.** The algorithm iterates on the full path of labor allocations $\{L_t^j\}$ until convergence. In each iteration, the dynamic module computes value functions and migration flows given current prices and wages, while the static module updates those prices and wages given the resulting labor allocations. Convergence is achieved when

$$\max_{j,t} \left| L_t^{j,\text{new}} - L_t^{j,\text{old}} \right| < \varepsilon,$$

with tolerance $\varepsilon = 10^{-3}$ in the baseline.

This iterative procedure jointly determines wages, prices, migration probabilities, and labor allocations such that households' forward-looking migration decisions are consistent with equilibrium prices in all regions and periods. It preserves the full regional income distribution—including the discrete agricultural mass and continuous non-agricultural tail—and correctly accounts for income-dependent migration costs and income persistence.

G. Additional quantitative results

Destination patterns. Beyond increasing overall mobility, the subsidy also changes *where* households migrate. Table G.8 decomposes average migration probabilities into moves toward rural and urban destinations, weighted by population and averaged over time. Under the baseline climate scenario, migration flows of low-income households are primarily directed toward rural regions, reflecting both affordability constraints and proximity to agricultural employment opportunities. Once the subsidy is introduced, these patterns shift markedly: low-income households' migration to rural destinations more than doubles (from 3.5% to 9.2%), indicating that the removal of liquidity constraints facilitates movement even within the rural economy. At the same time, their urban migration rate remains nearly unchanged, suggesting that the policy does not disproportionately encourage rural–urban exodus. Middle-income households also exhibit modest increases in rural mobility, whereas high-income households—who were largely unconstrained—show little change or slightly lower overall migration. Together, these findings confirm that the subsidy primarily enhances *adaptive migration within and between rural areas*, rather than triggering large-scale rural–urban shifts.

TABLE G.8
Effect of the migration subsidy on rural and urban destination choices (climate path)

Income group	Migration to rural regions (%)	Migration to urban regions (%)
Low income ($\leq$10k)	3.54 $\rightarrow$ 9.20	1.87 $\rightarrow$ 1.89
Middle income (10–20k)	2.07 $\rightarrow$ 2.53	4.32 $\rightarrow$ 2.88
High income (>20k)	2.87 $\rightarrow$ 2.54	8.90 $\rightarrow$ 7.09

Notes: Each entry compares the baseline and subsidized regimes under the same climate path. "Migration to rural (urban) regions" reports the population- and time-weighted average probability of migrating to destinations classified as rural (urban) based on whether their agricultural employment share λ_a lies above or below the national median. Arrows ($\rightarrow$) indicate the change from the baseline to the subsidized regime.

Additional references

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